

# BFS Traversal

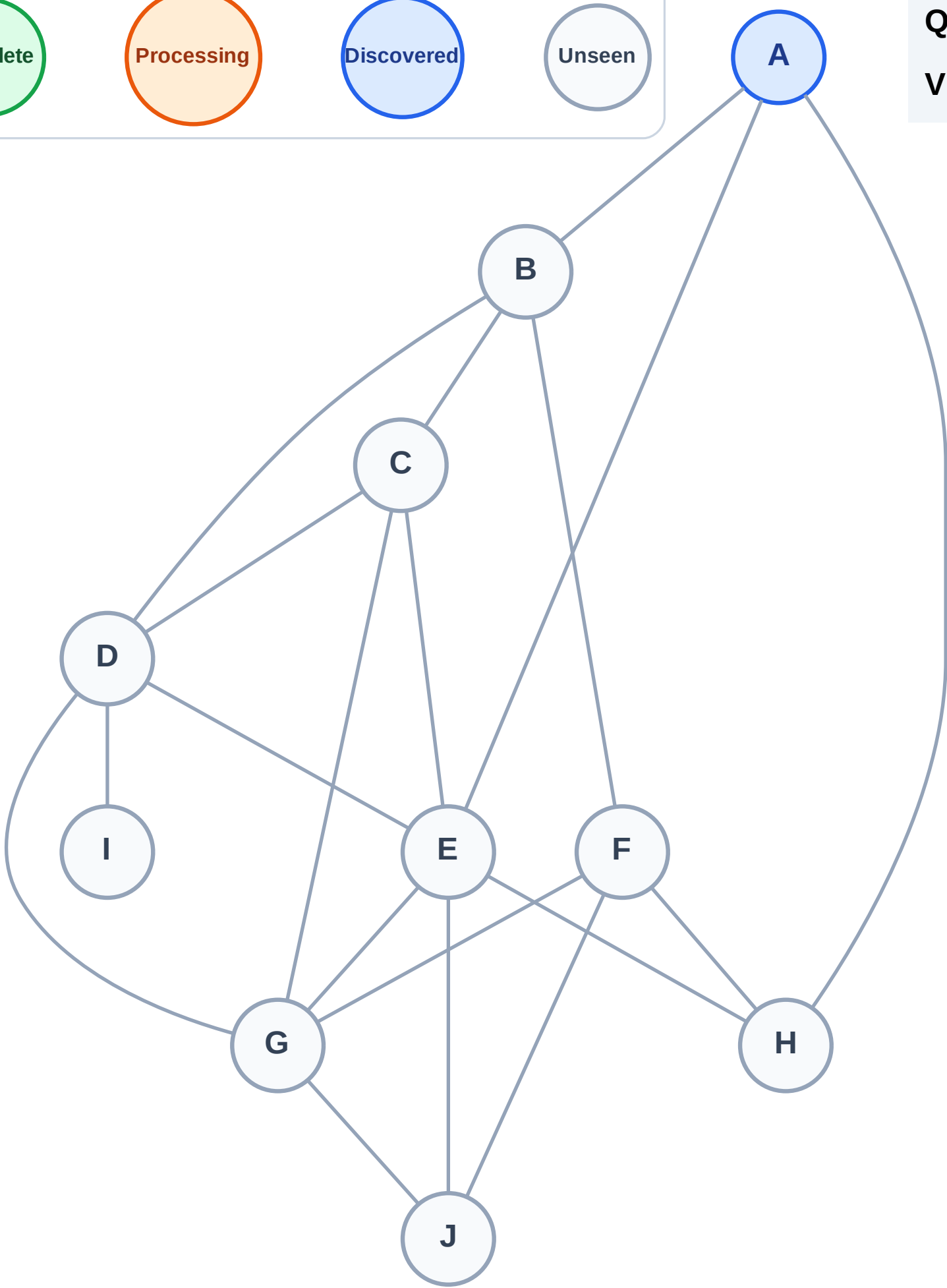
Step 1: Start at A

## Legend



Queue: A

Visited: (none)



# BFS Traversal

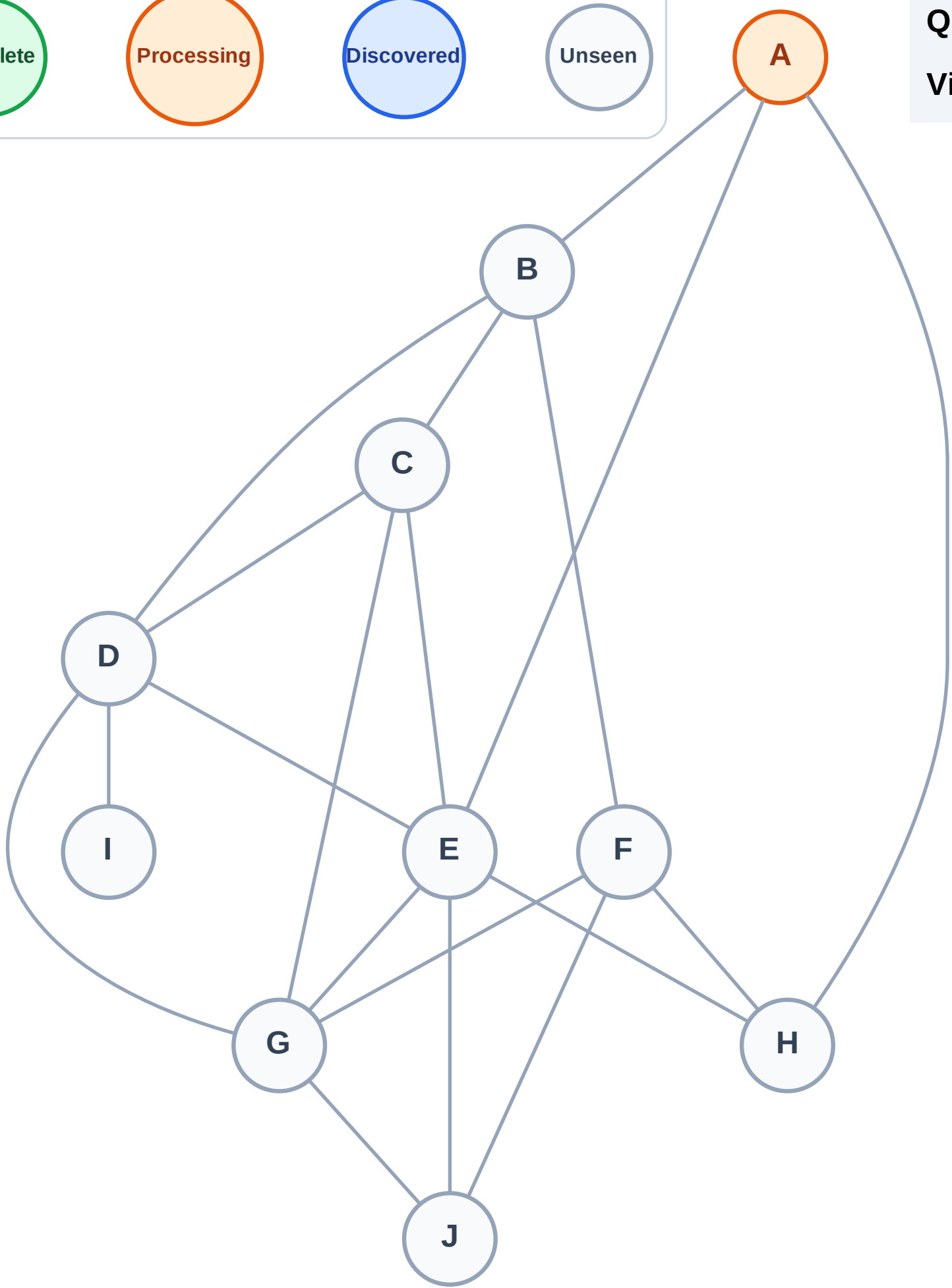
Step 2: Process node A

Legend



Queue: (empty)

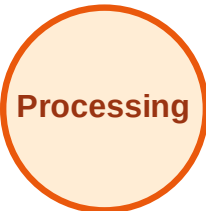
Visited: (none)



# BFS Traversal

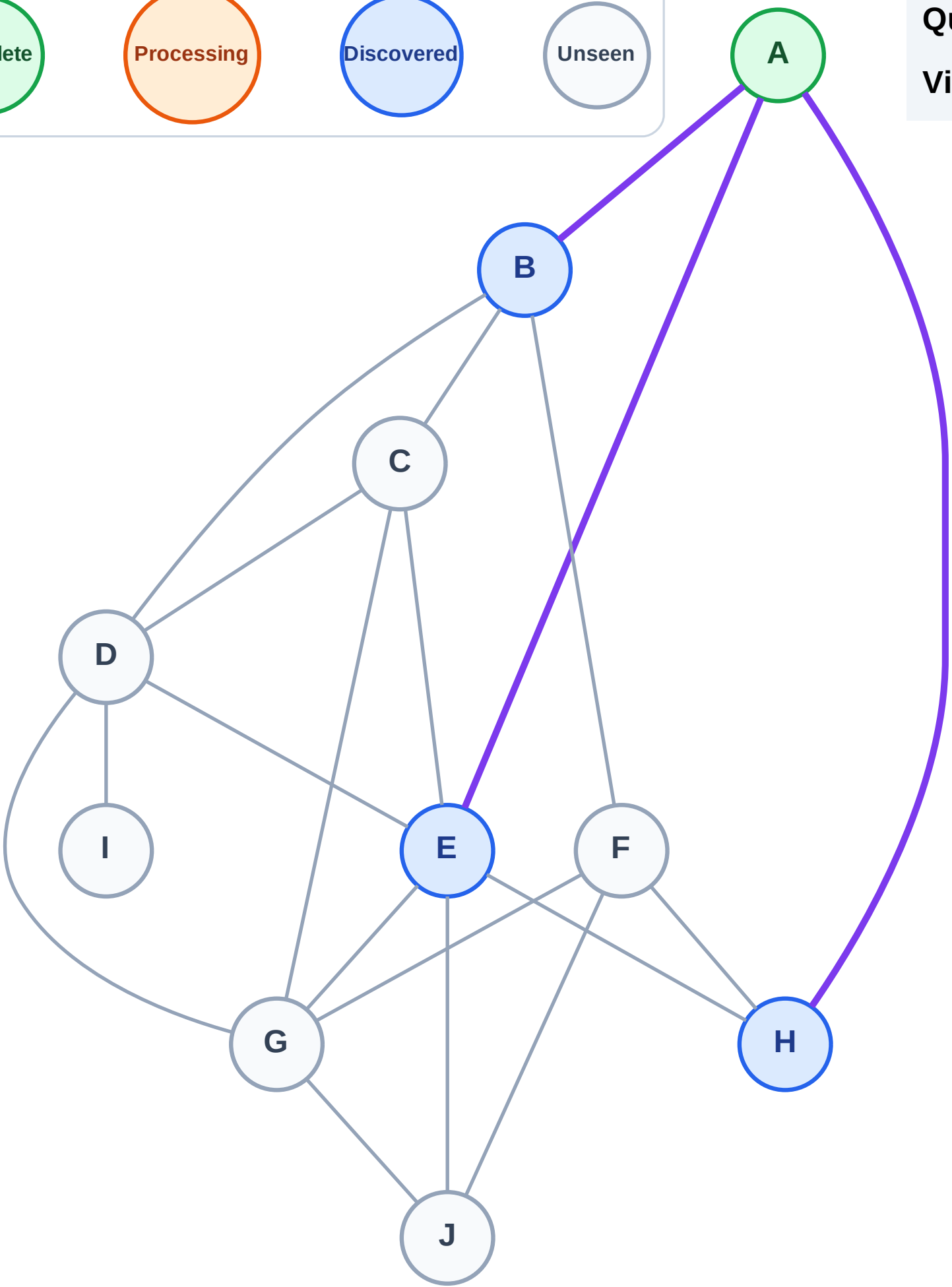
Step 3: Discover B, E, H from A

## Legend



Queue: B → E → H

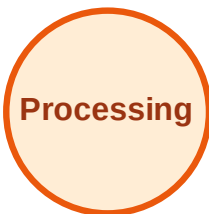
Visited: A



# BFS Traversal

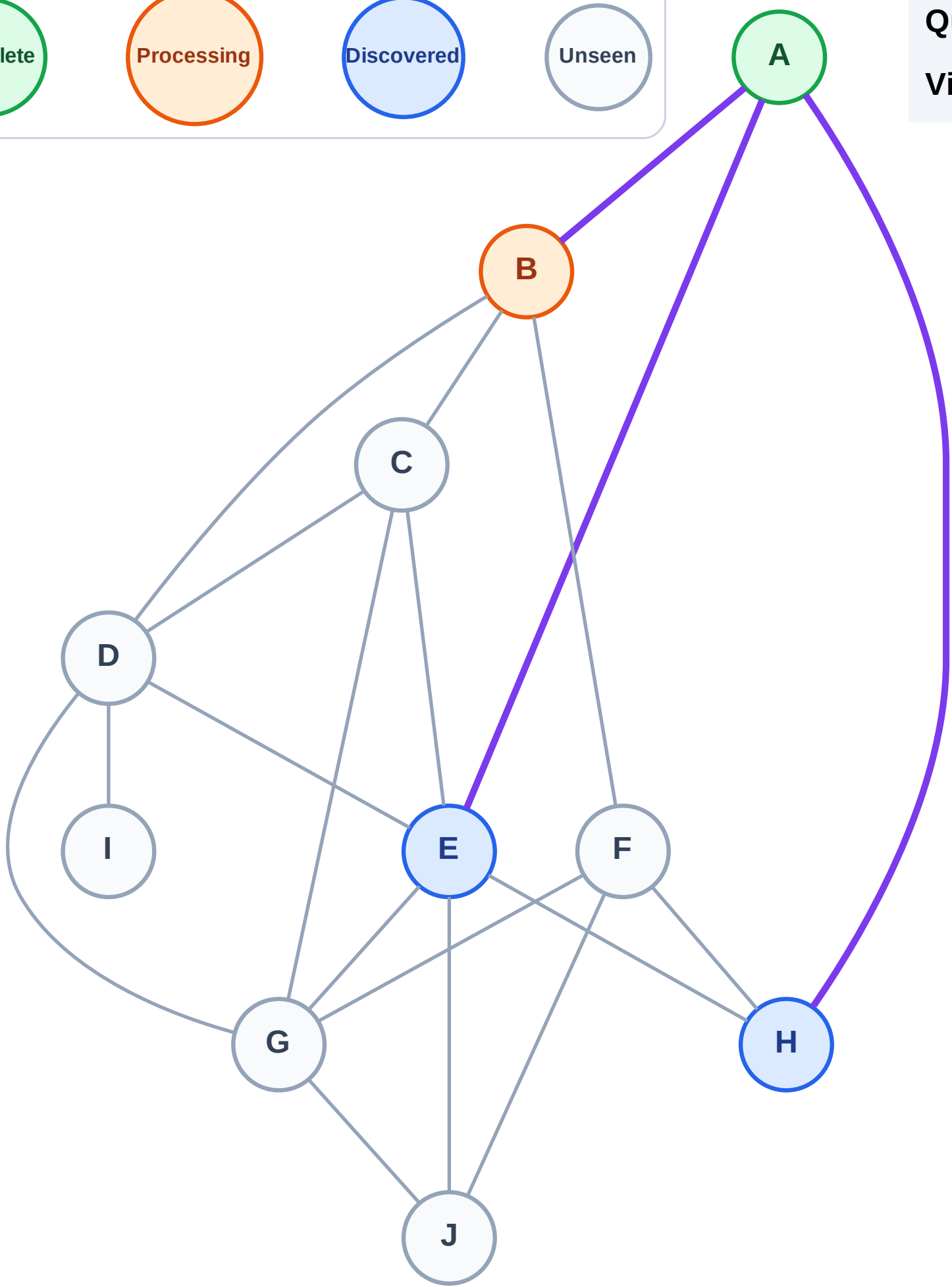
Step 4: Process node B

## Legend



Queue: E → H

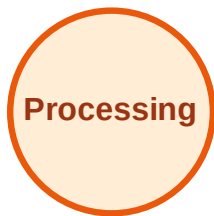
Visited: A



# BFS Traversal

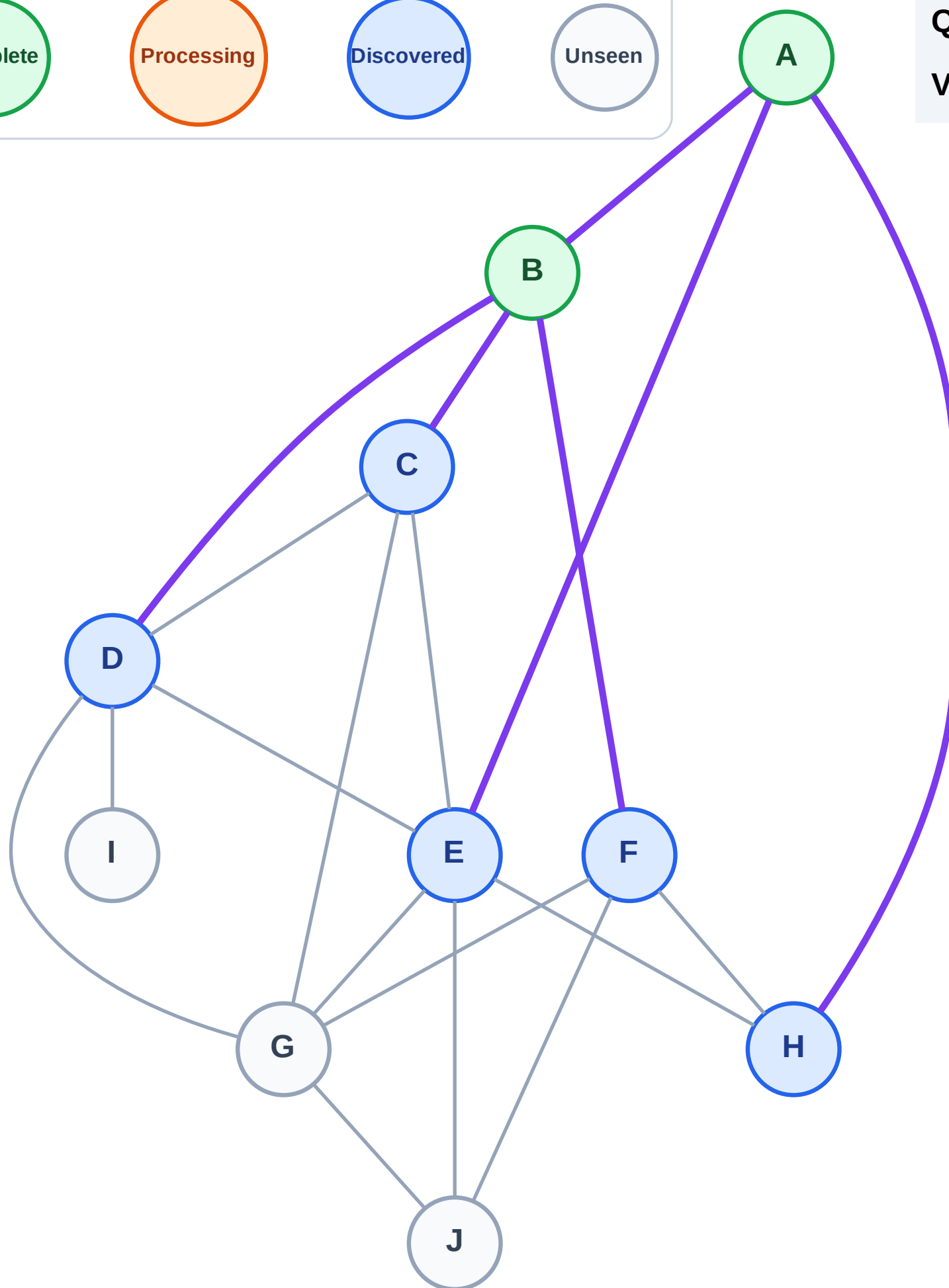
Step 5: Discover C, D, F from B

## Legend



Queue: E → H → C → D → F

Visited: A, B



# BFS Traversal

Step 6: Process node E

Legend

Complete

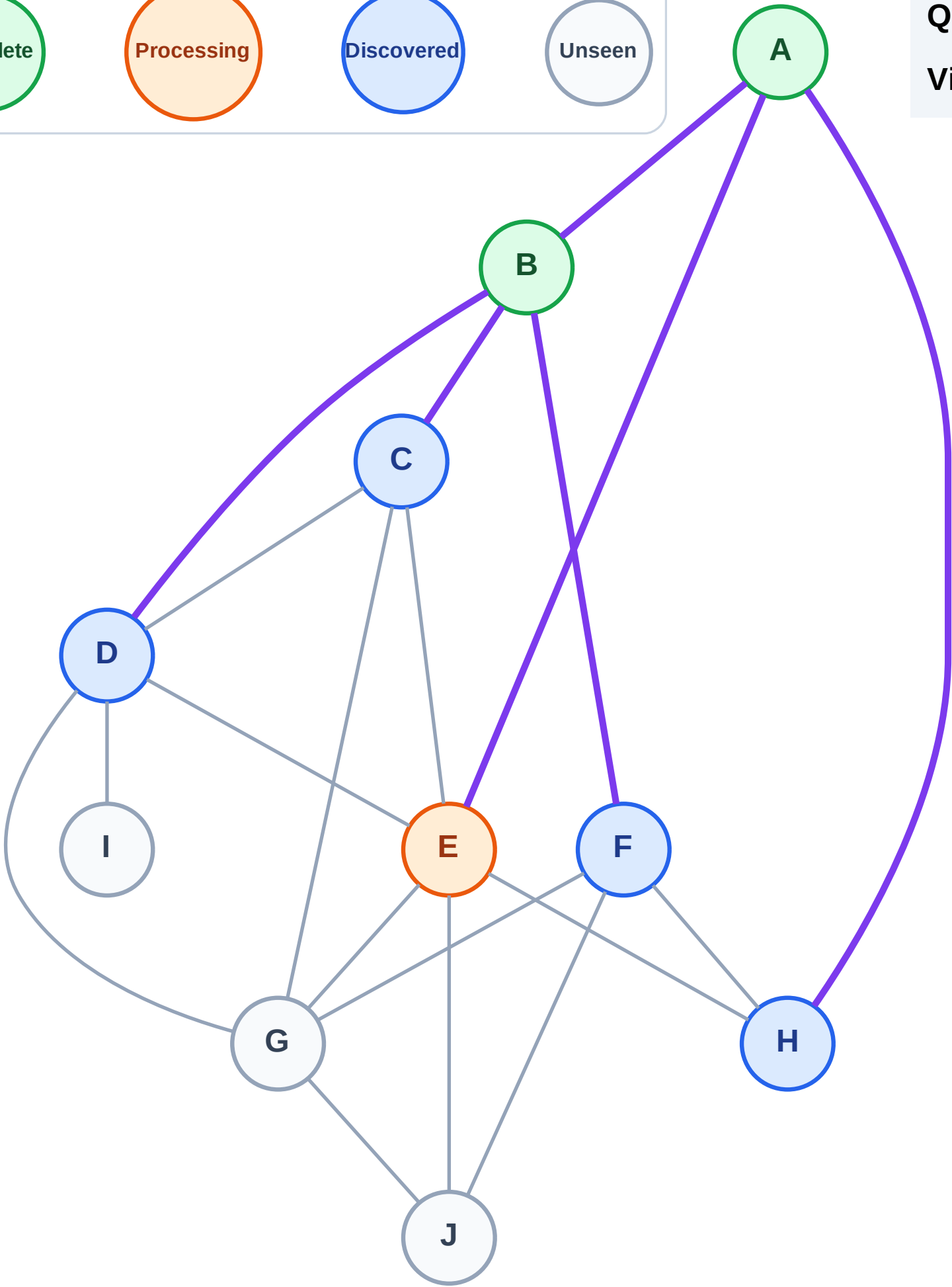
Processing

Discovered

Unseen

Queue: H → C → D → F

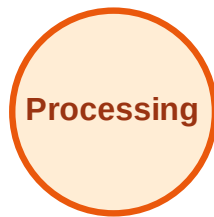
Visited: A, B



# BFS Traversal

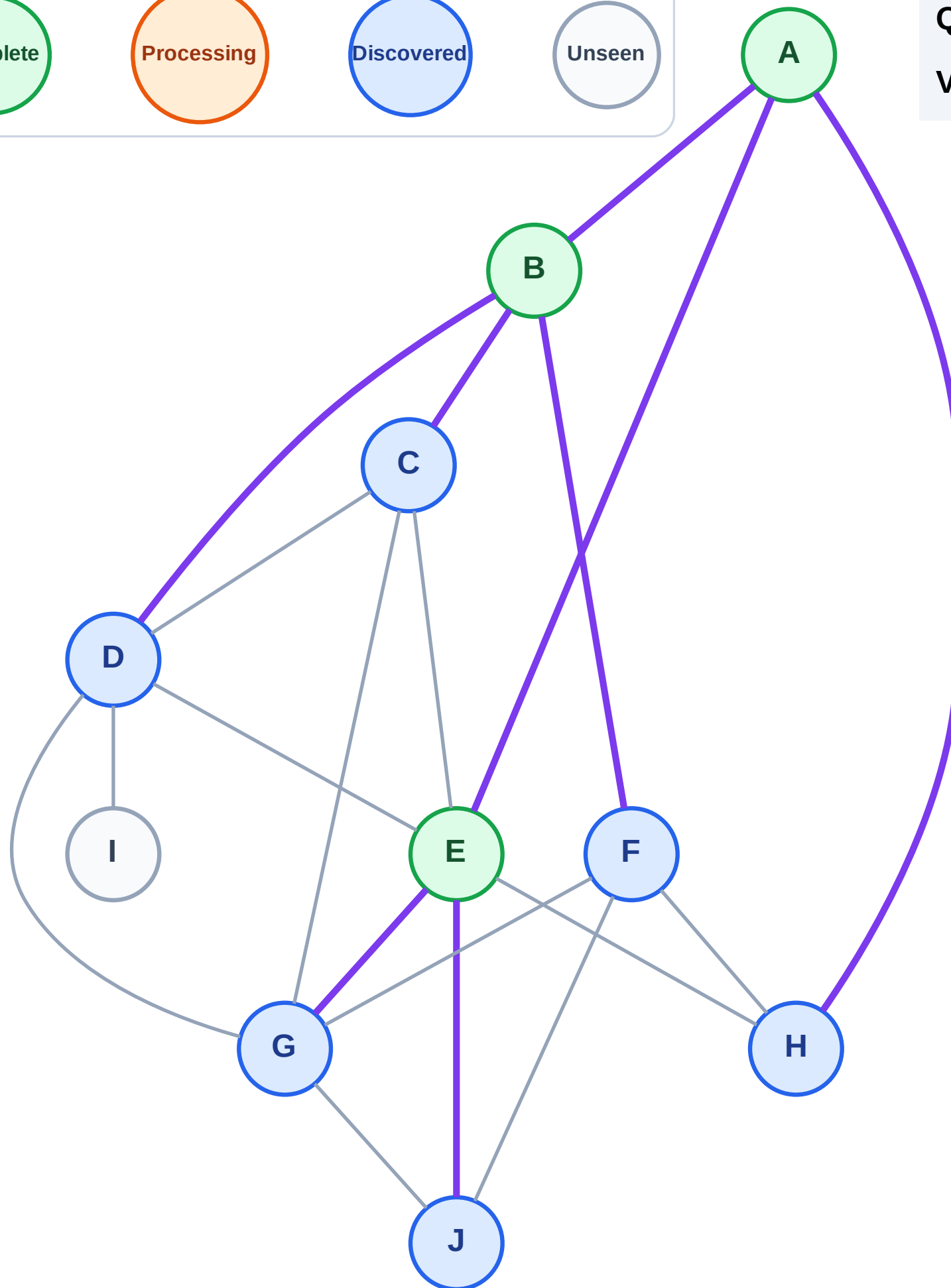
Step 7: Discover G, J from E

## Legend



Queue: H → C → D → F → G → J

Visited: A, B, E



# BFS Traversal

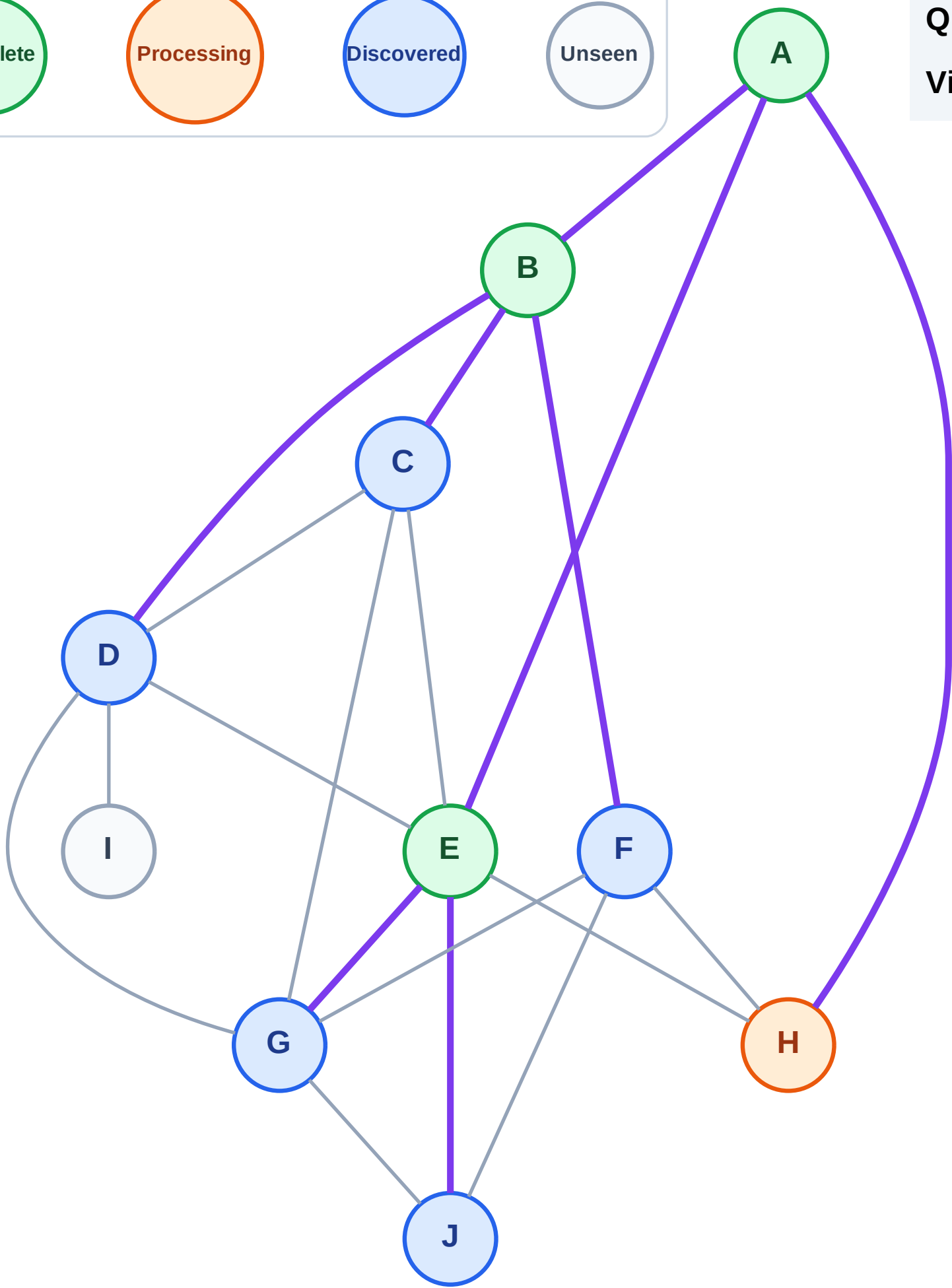
Step 8: Process node H

## Legend



Queue: C → D → F → G → J

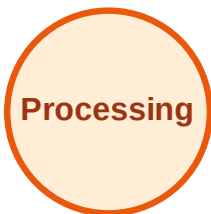
Visited: A, B, E



# BFS Traversal

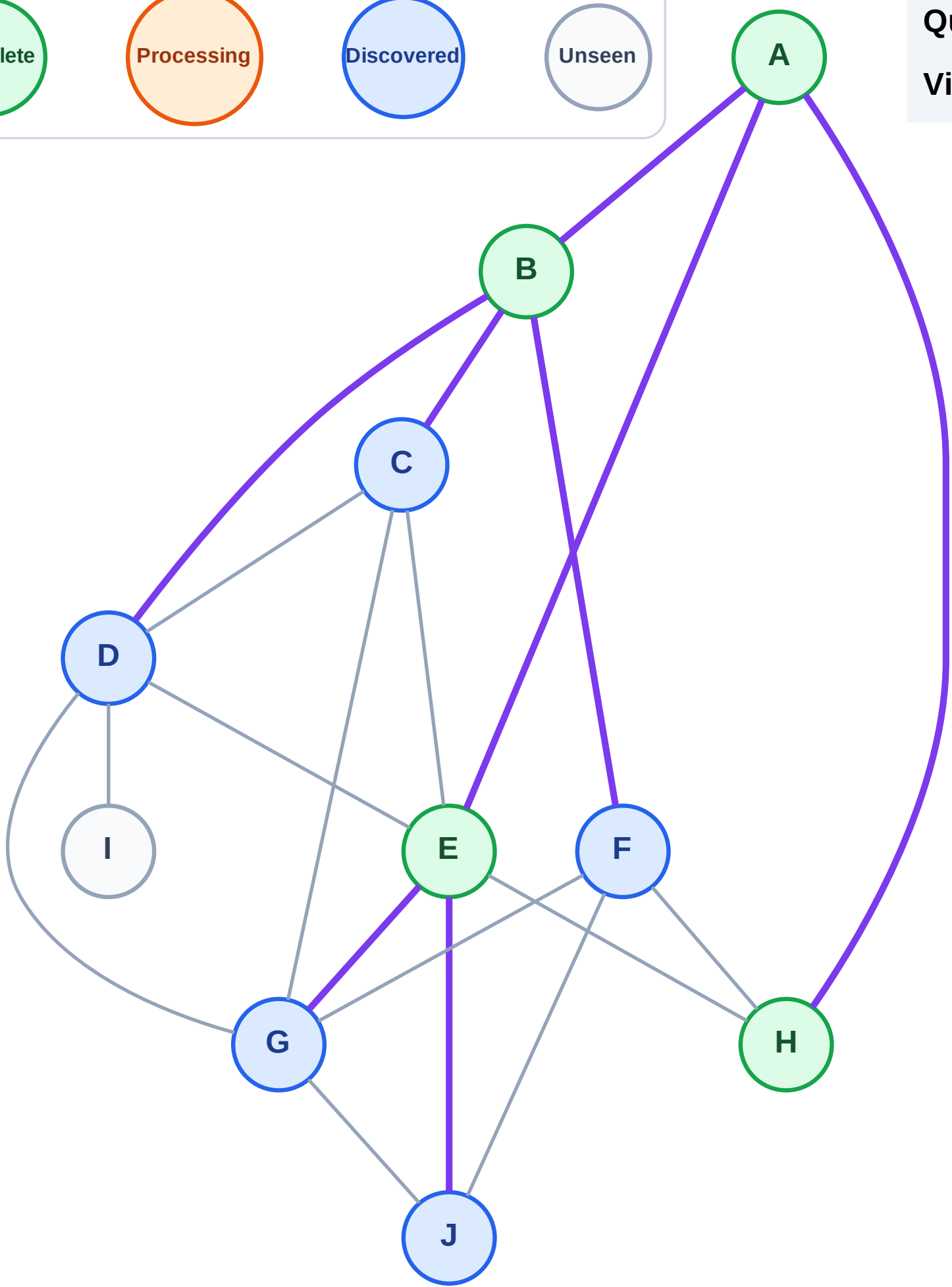
Step 9: No new neighbors from H

## Legend



Queue: C → D → F → G → J

Visited: A, B, E, H



# BFS Traversal

Step 10: Process node C

## Legend

Complete

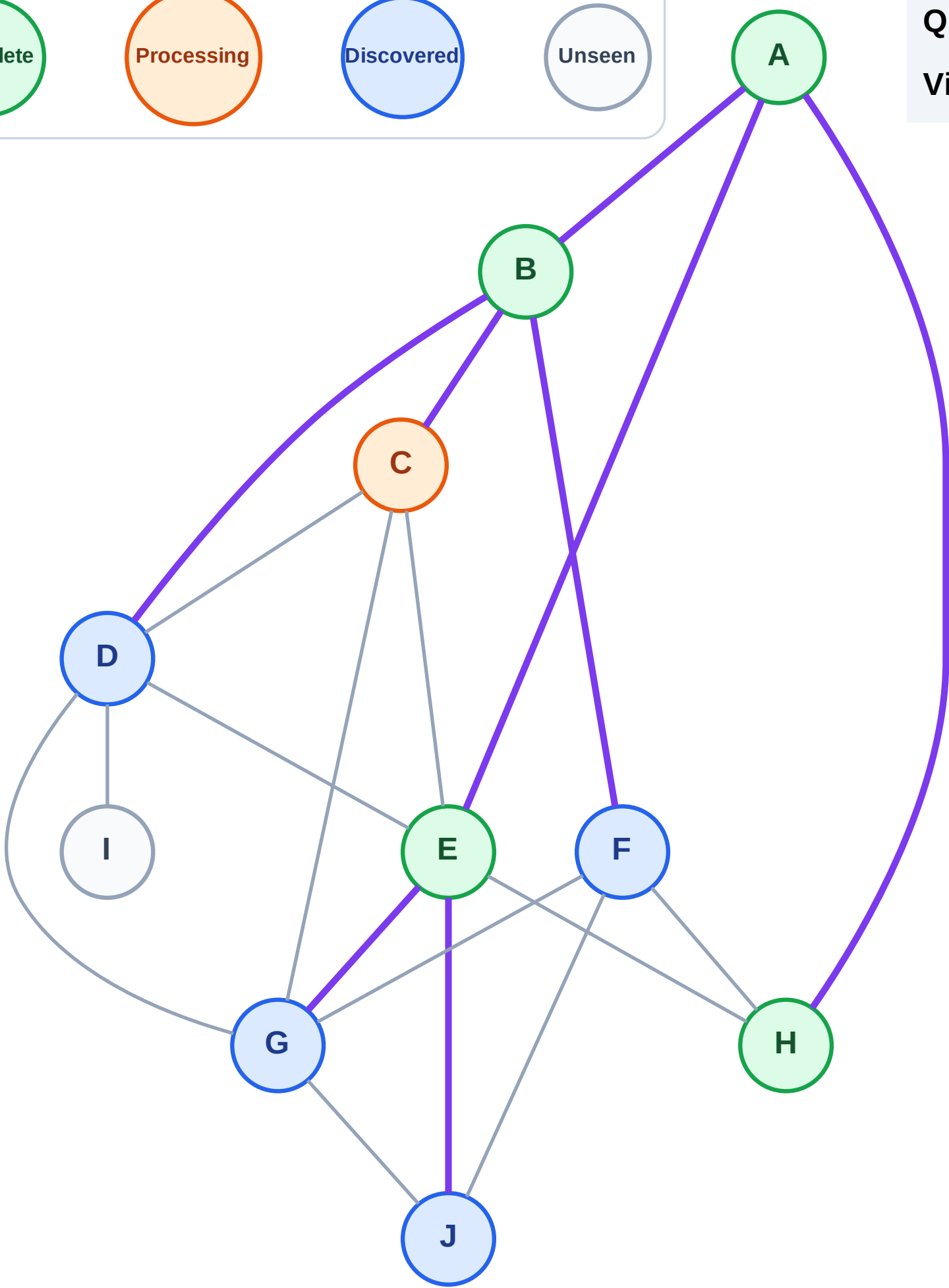
Processing

Discovered

Unseen

Queue: D → F → G → J

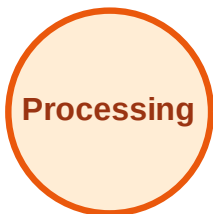
Visited: A, B, E, H



# BFS Traversal

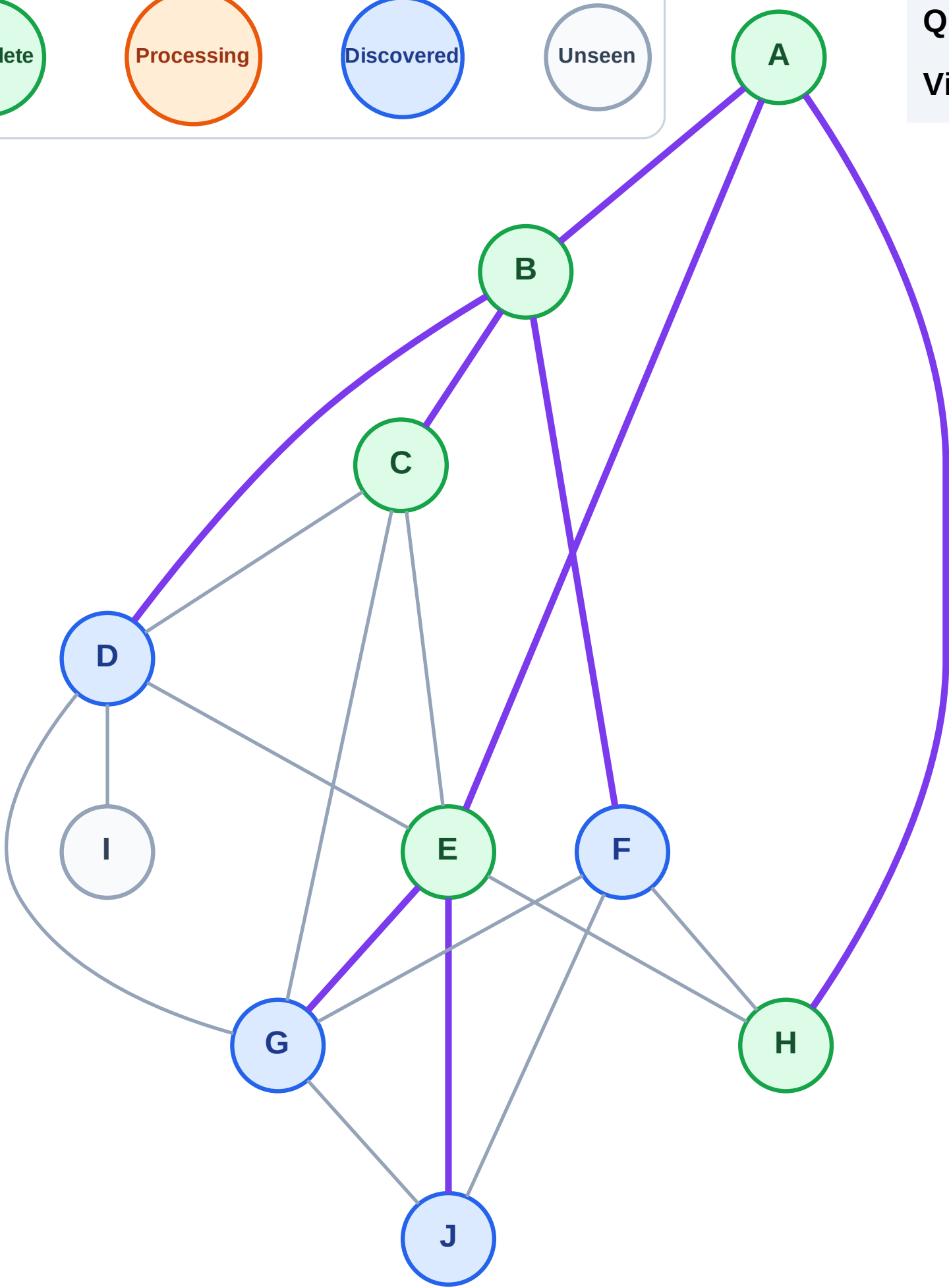
Step 11: No new neighbors from C

## Legend



Queue: D → F → G → J

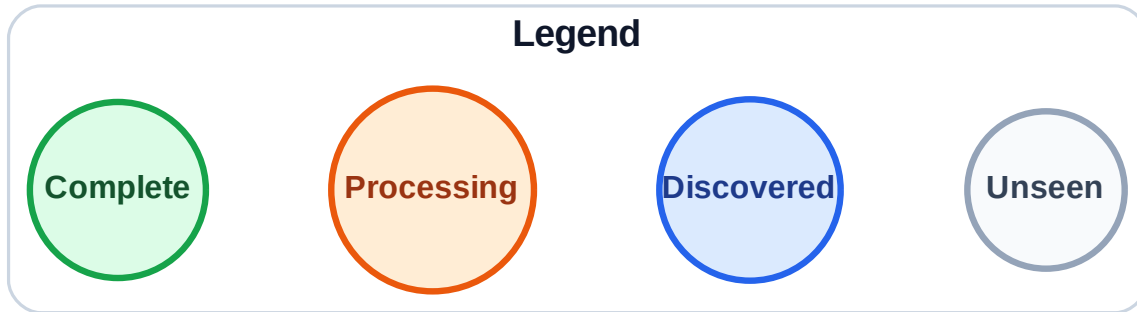
Visited: A, B, C, E, H



## BFS Traversal

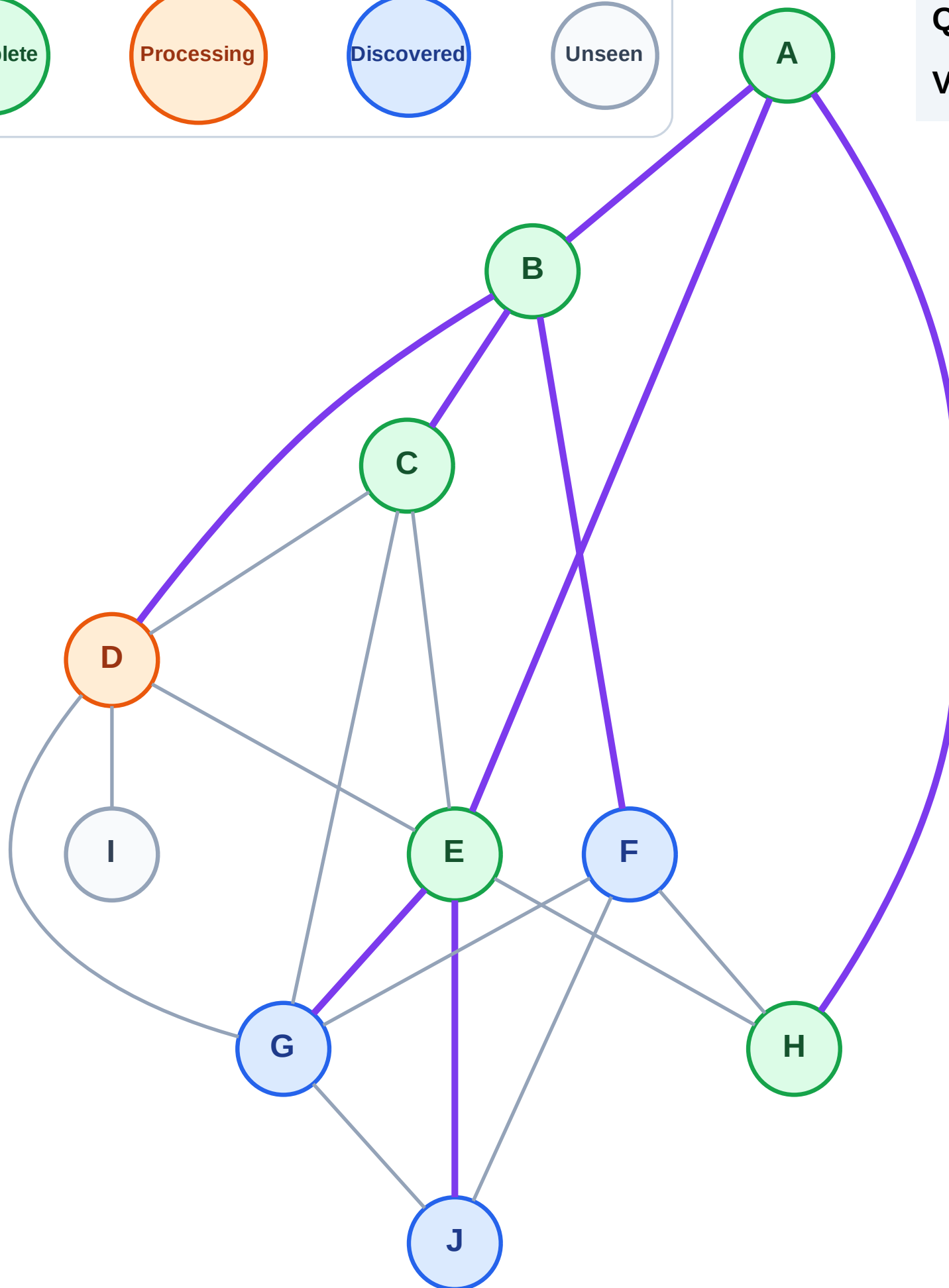
Step 12: Process node D

## Step 12: Process node D



**Queue: F → G → J**

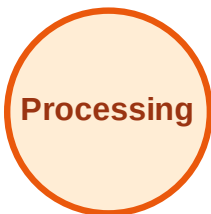
**Visited: A, B, C, E, H**



# BFS Traversal

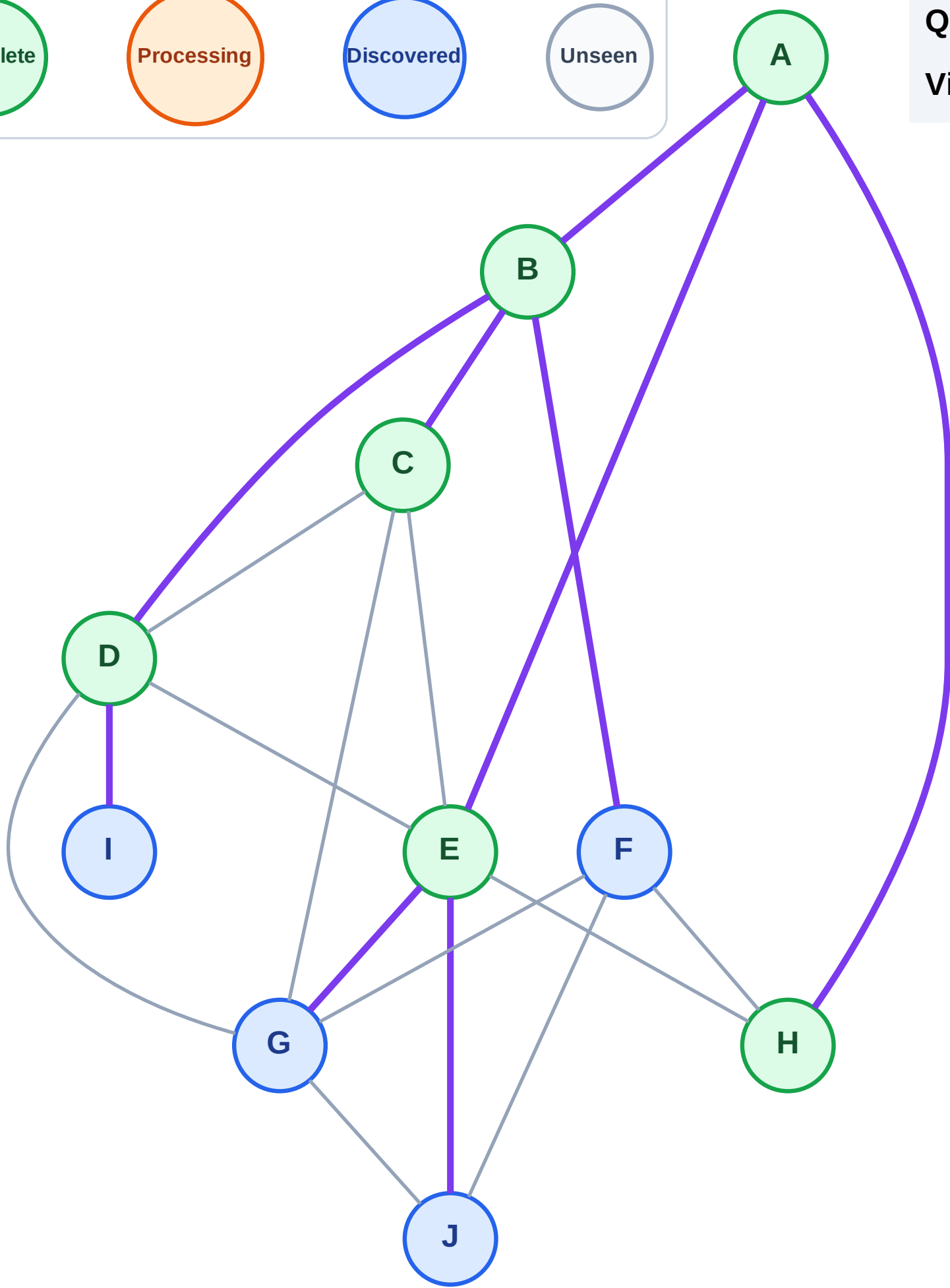
Step 13: Discover I from D

## Legend



Queue: F → G → J → I

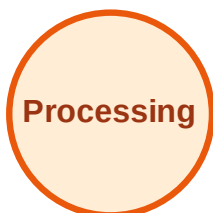
Visited: A, B, C, D, E, H



# BFS Traversal

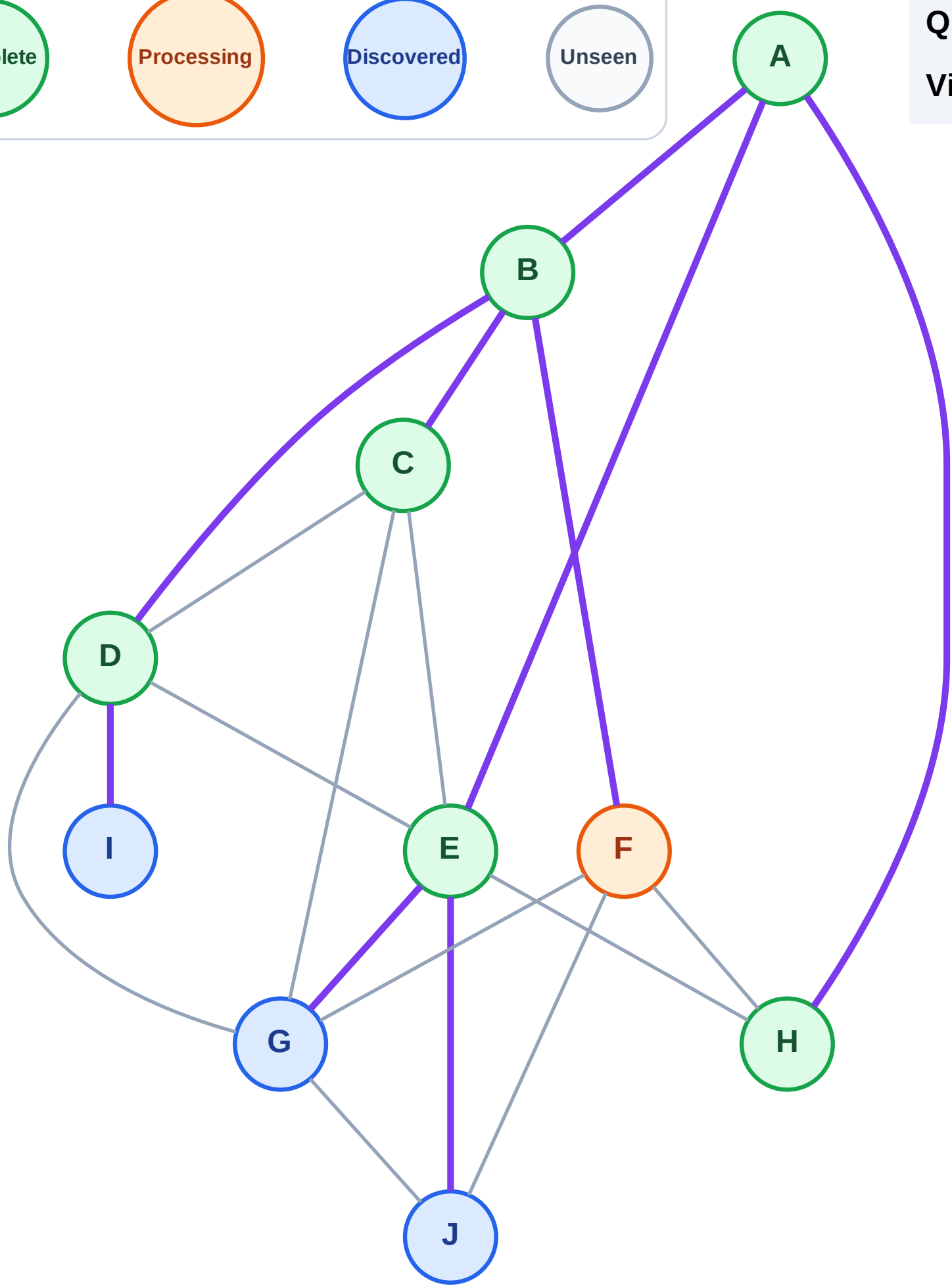
Step 14: Process node F

## Legend



Queue: G → J → I

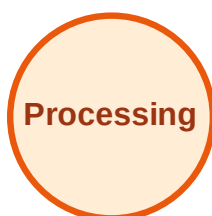
Visited: A, B, C, D, E, H



# BFS Traversal

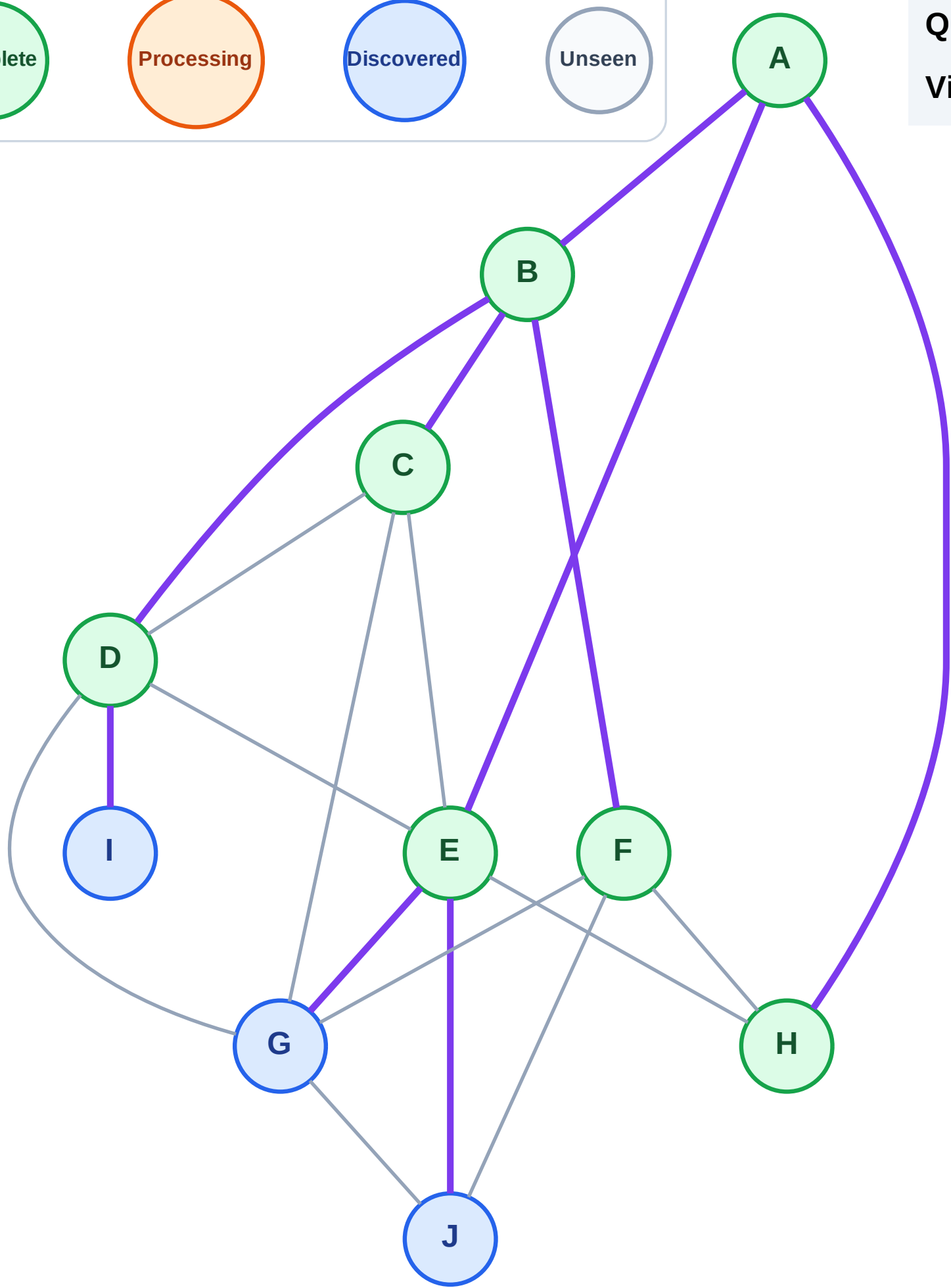
Step 15: No new neighbors from F

## Legend

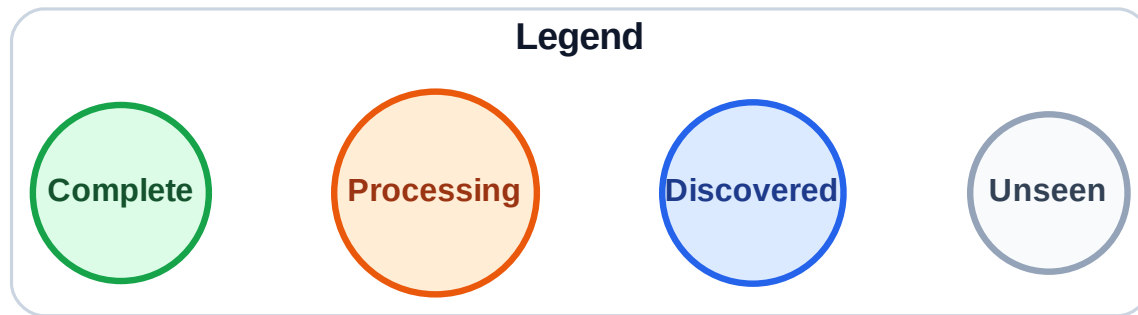


Queue: G → J → I

Visited: A, B, C, D, E, F, H

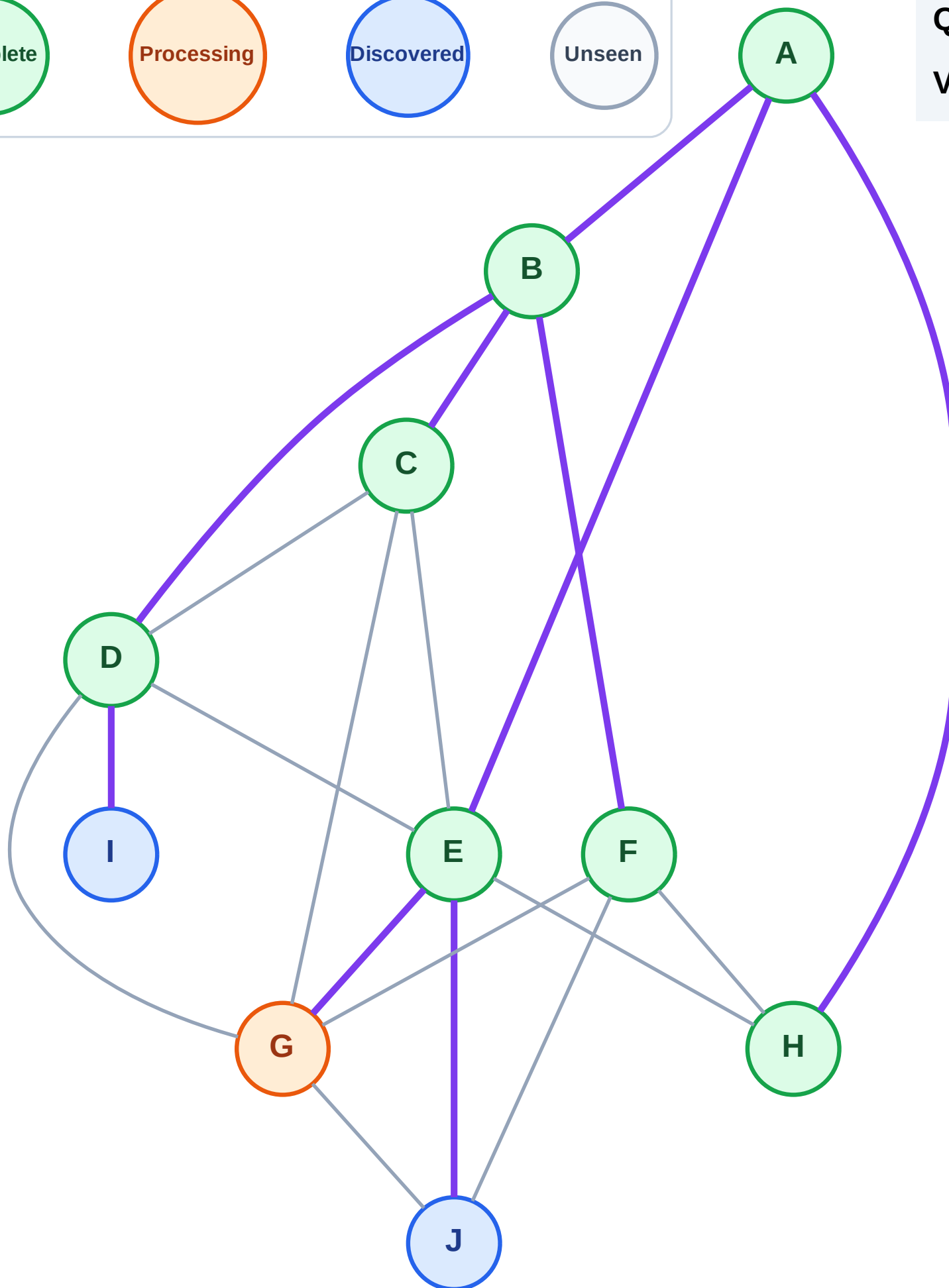


## Step 16: Process node G



**Queue: J → I**

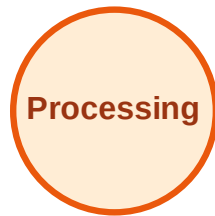
**Visited: A, B, C, D, E, F, H**



# BFS Traversal

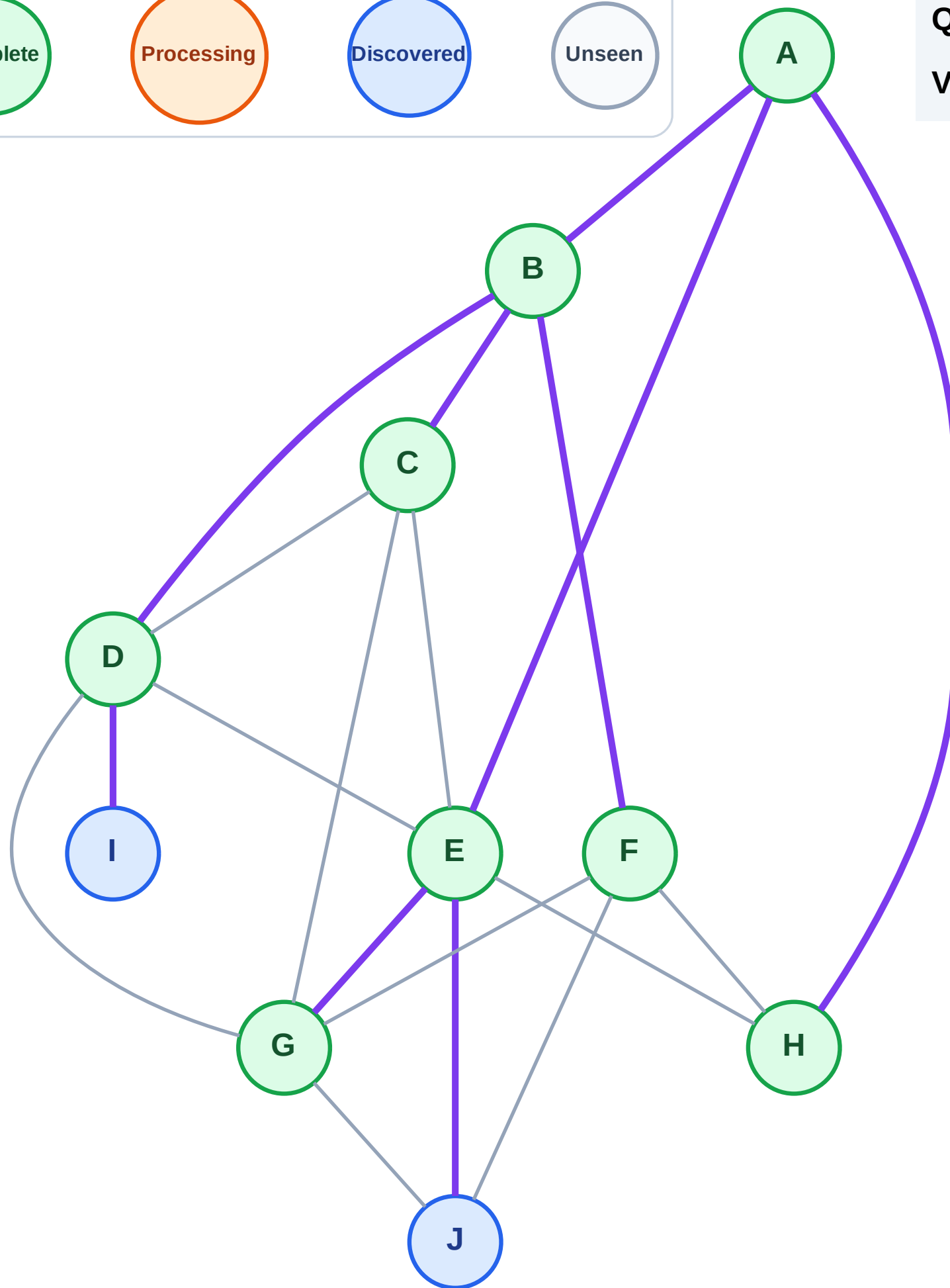
Step 17: No new neighbors from G

## Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



# BFS Traversal

Step 18: Process node J

## Legend

Complete

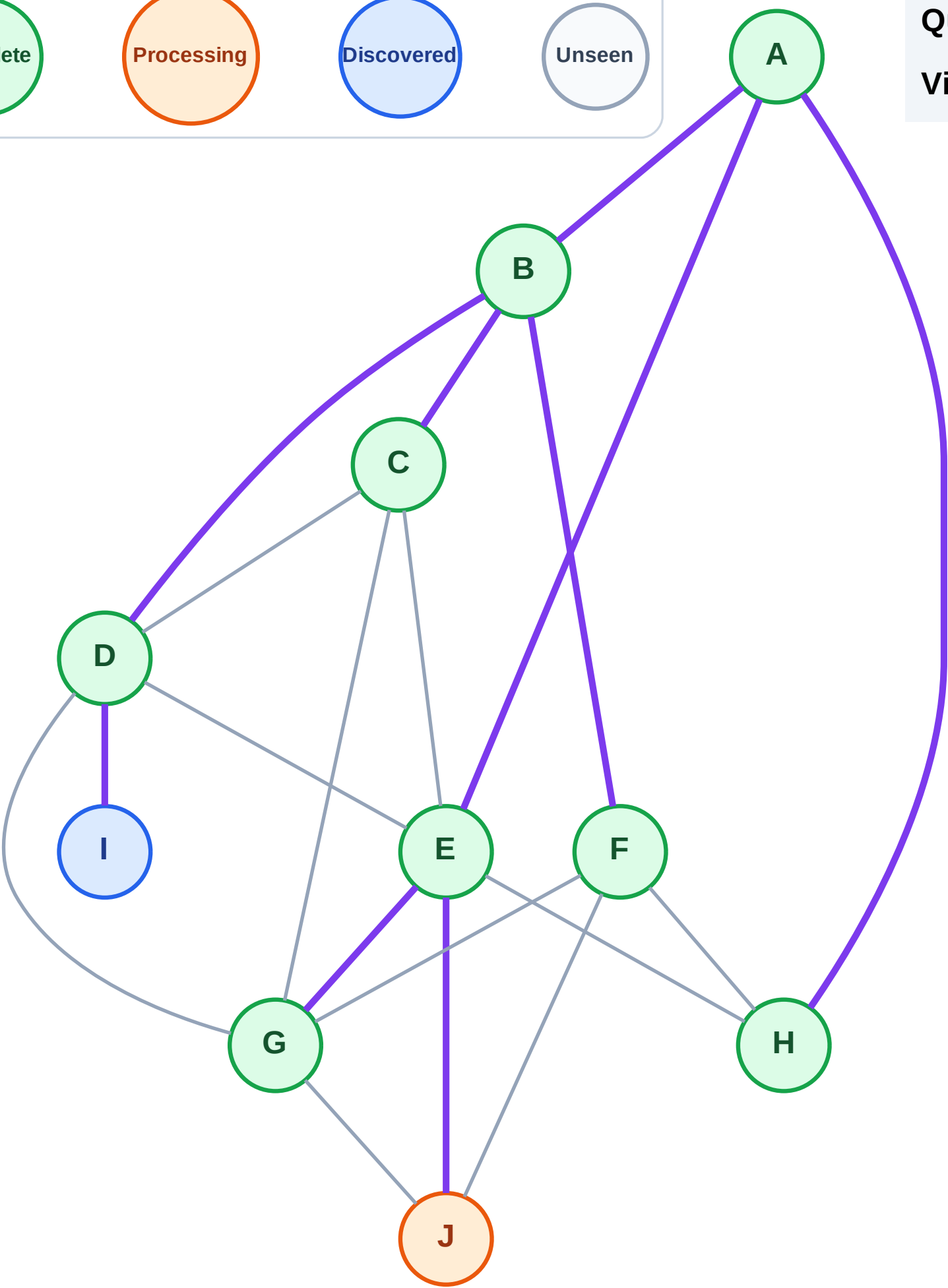
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, F, G, H



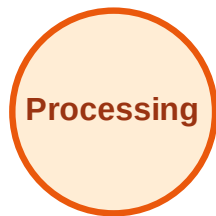
# BFS Traversal

Step 19: No new neighbors from J

## Legend



Complete



Processing



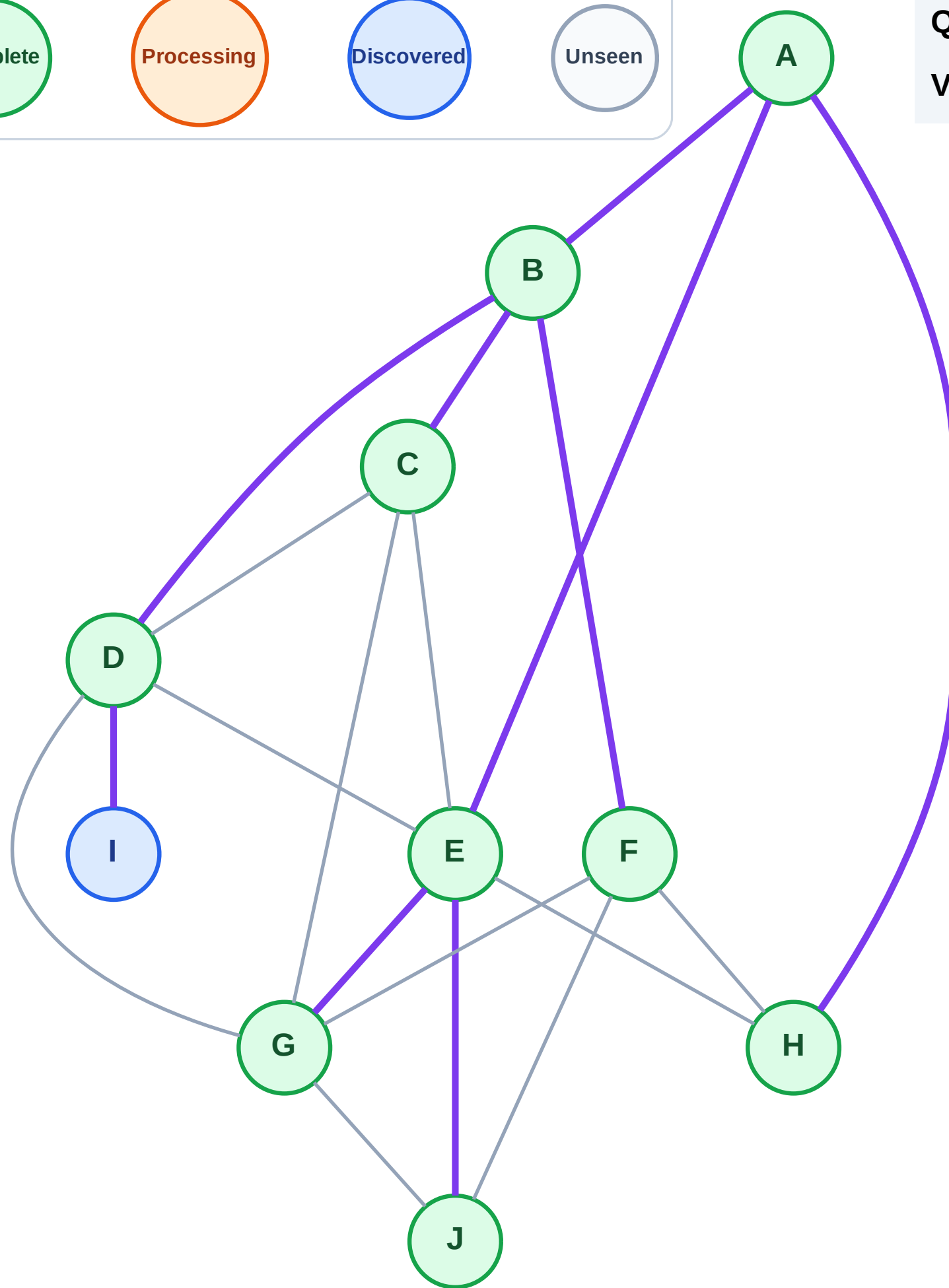
Discovered



Unseen

Queue: I

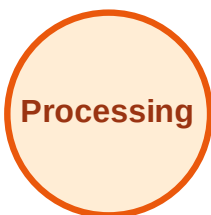
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

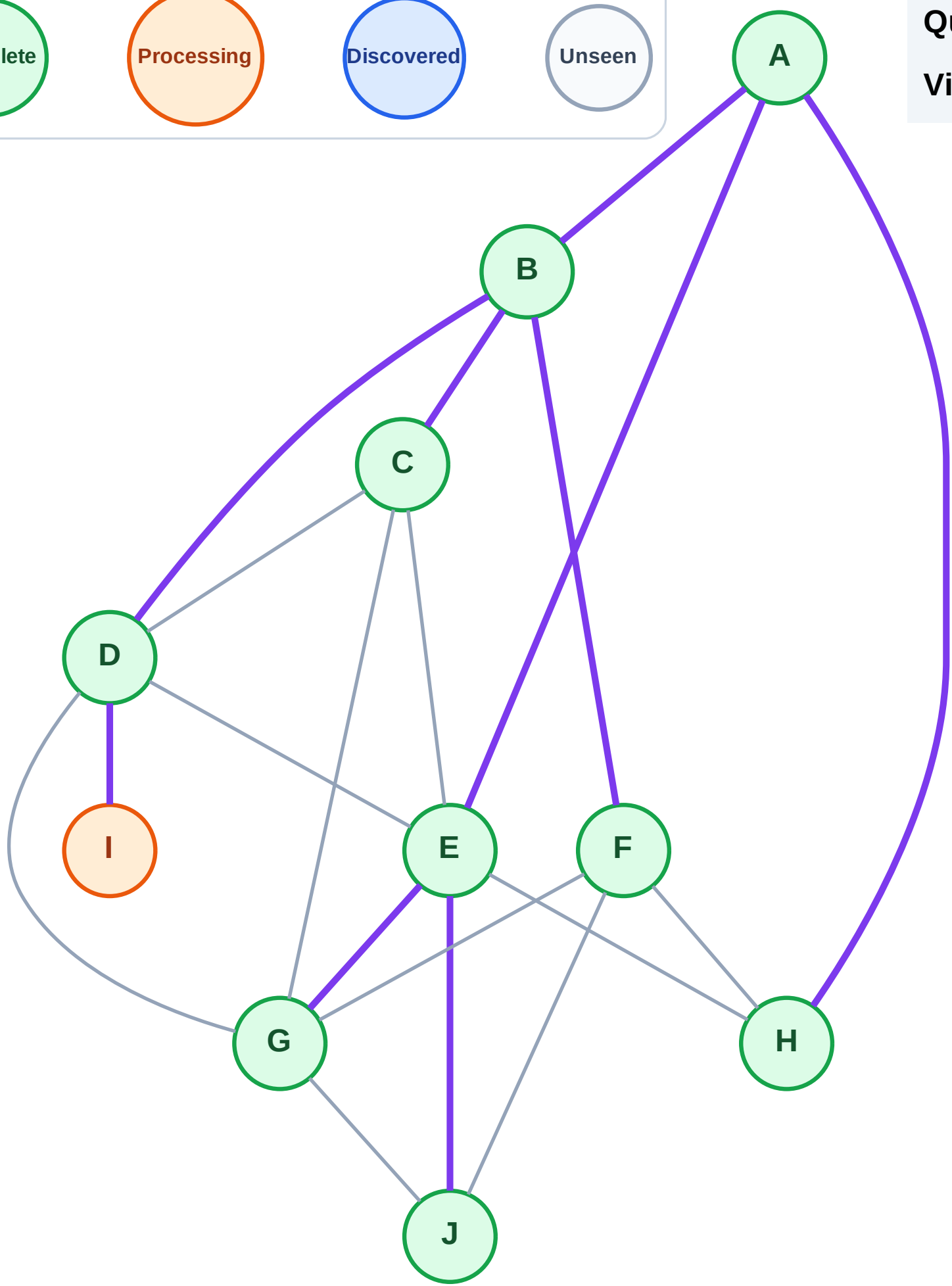
Step 20: Process node I

## Legend



Queue: (empty)

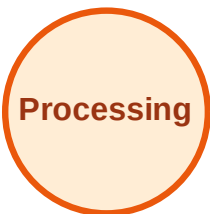
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

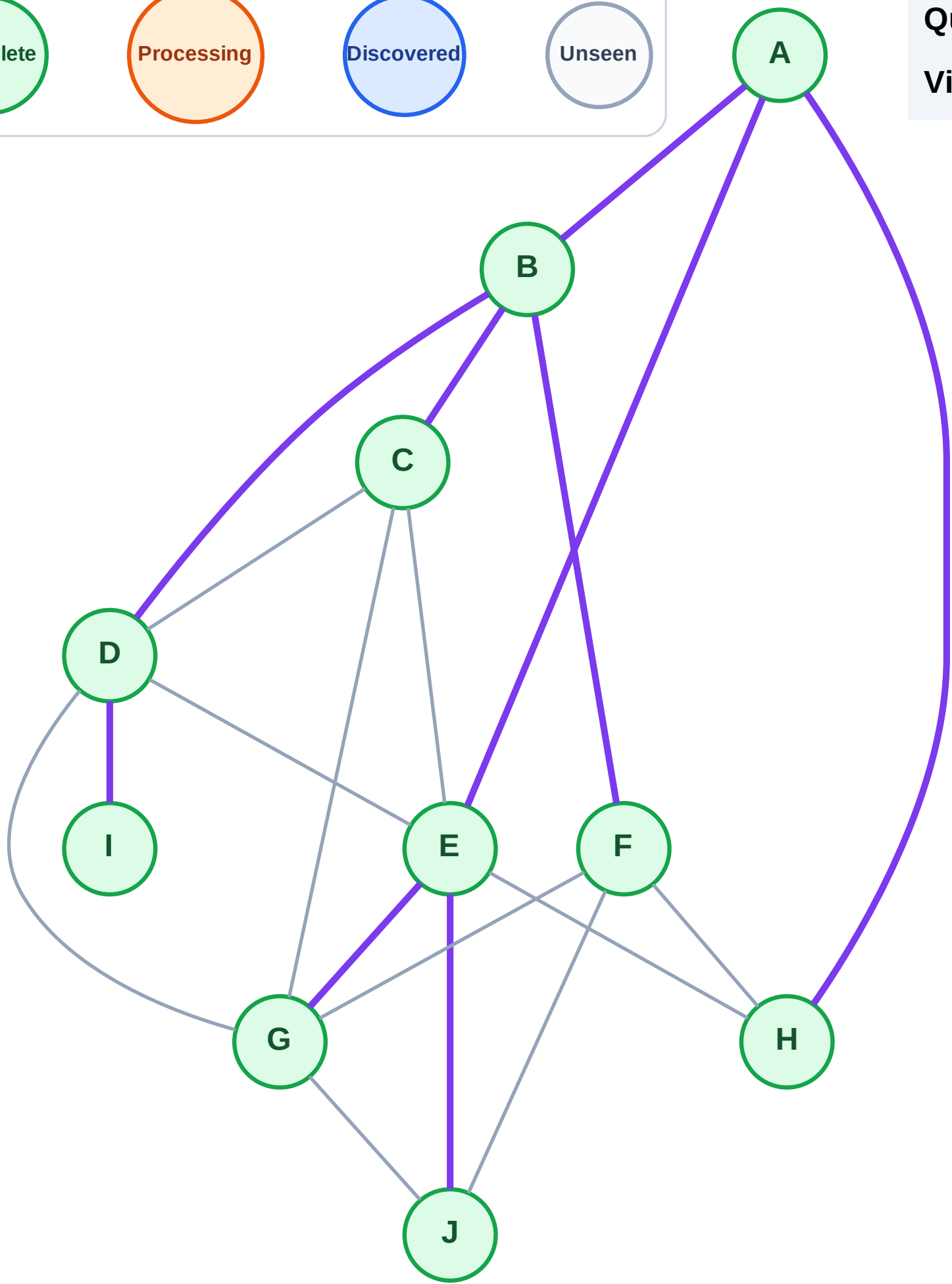
Step 21: No new neighbors from I

## Legend



Queue: (empty)

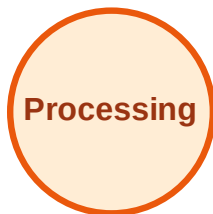
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

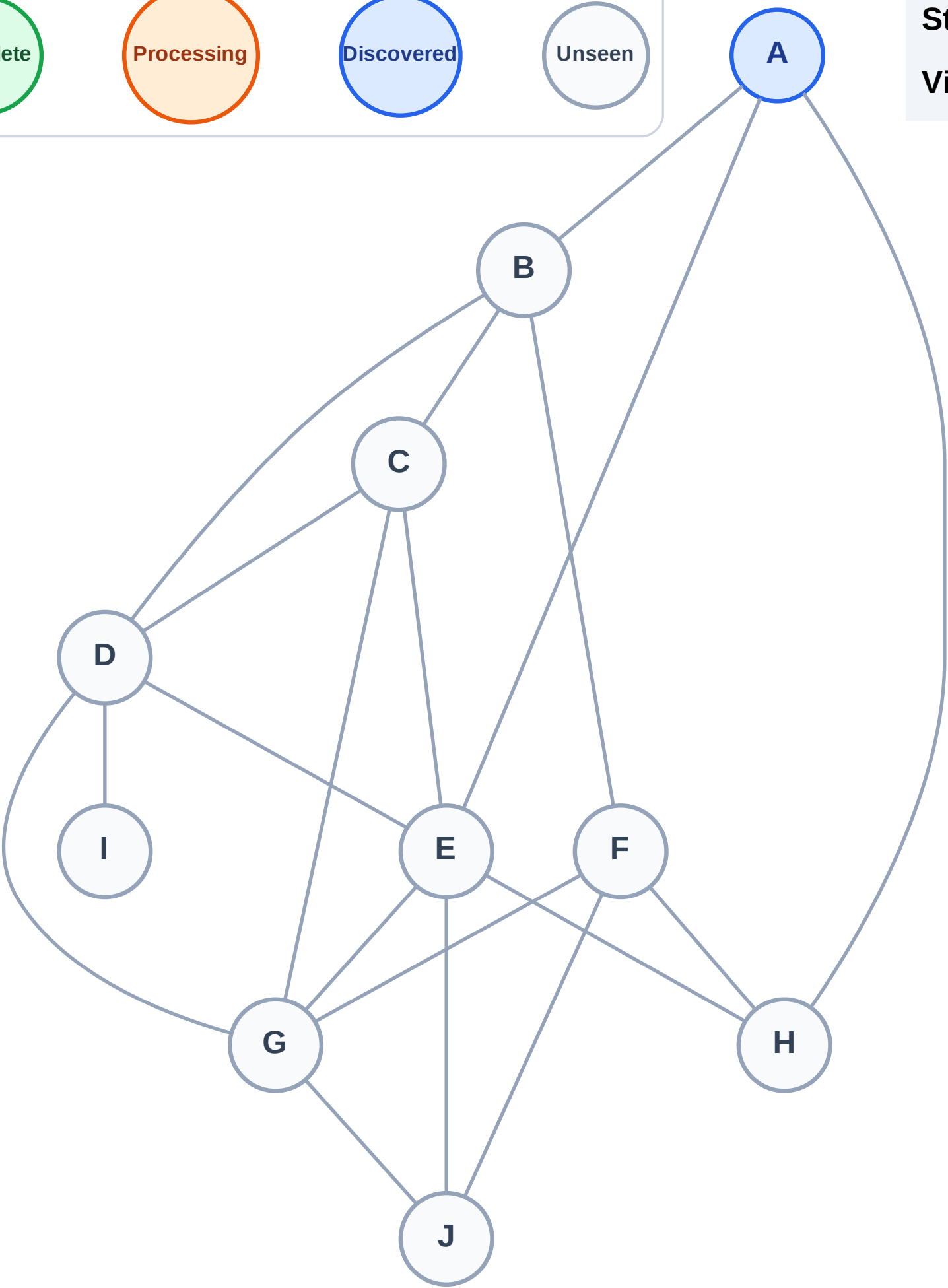
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

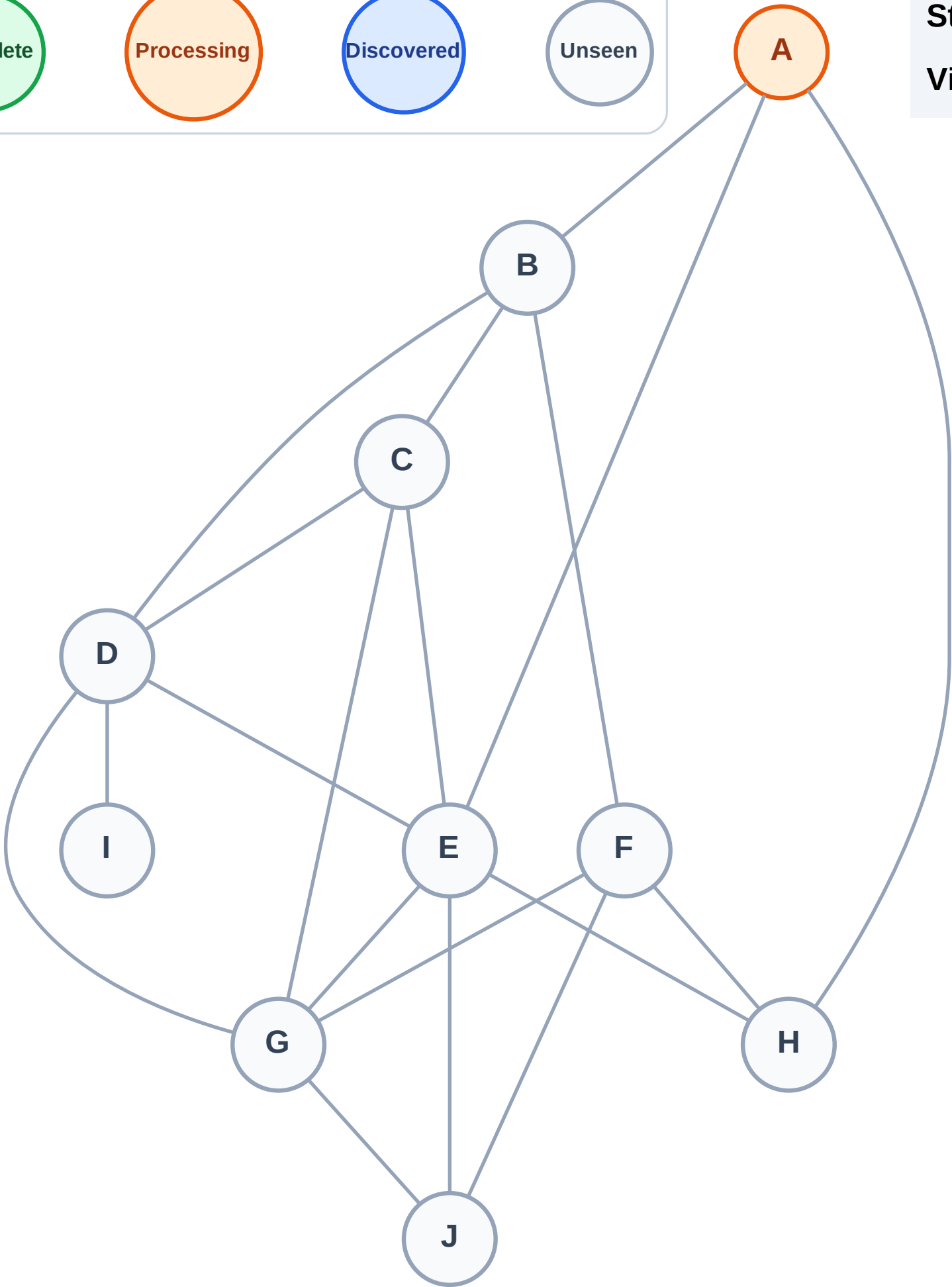
Complete

Processing

Discovered

Unseen

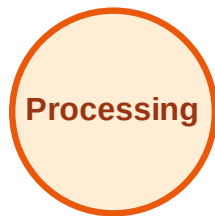
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

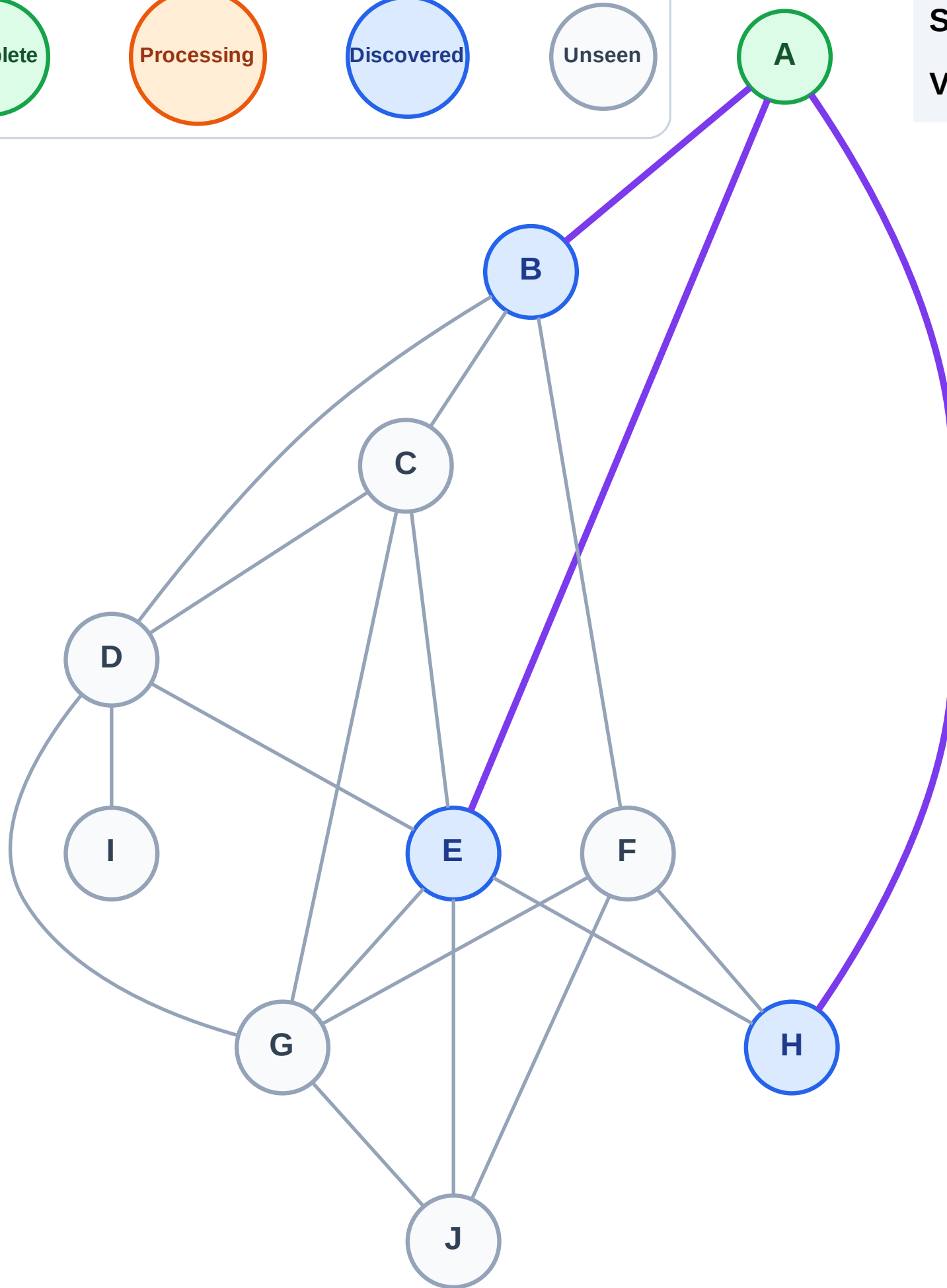
Step 3: Discover H, E, B from A

## Legend



Stack (top at right): H | E | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

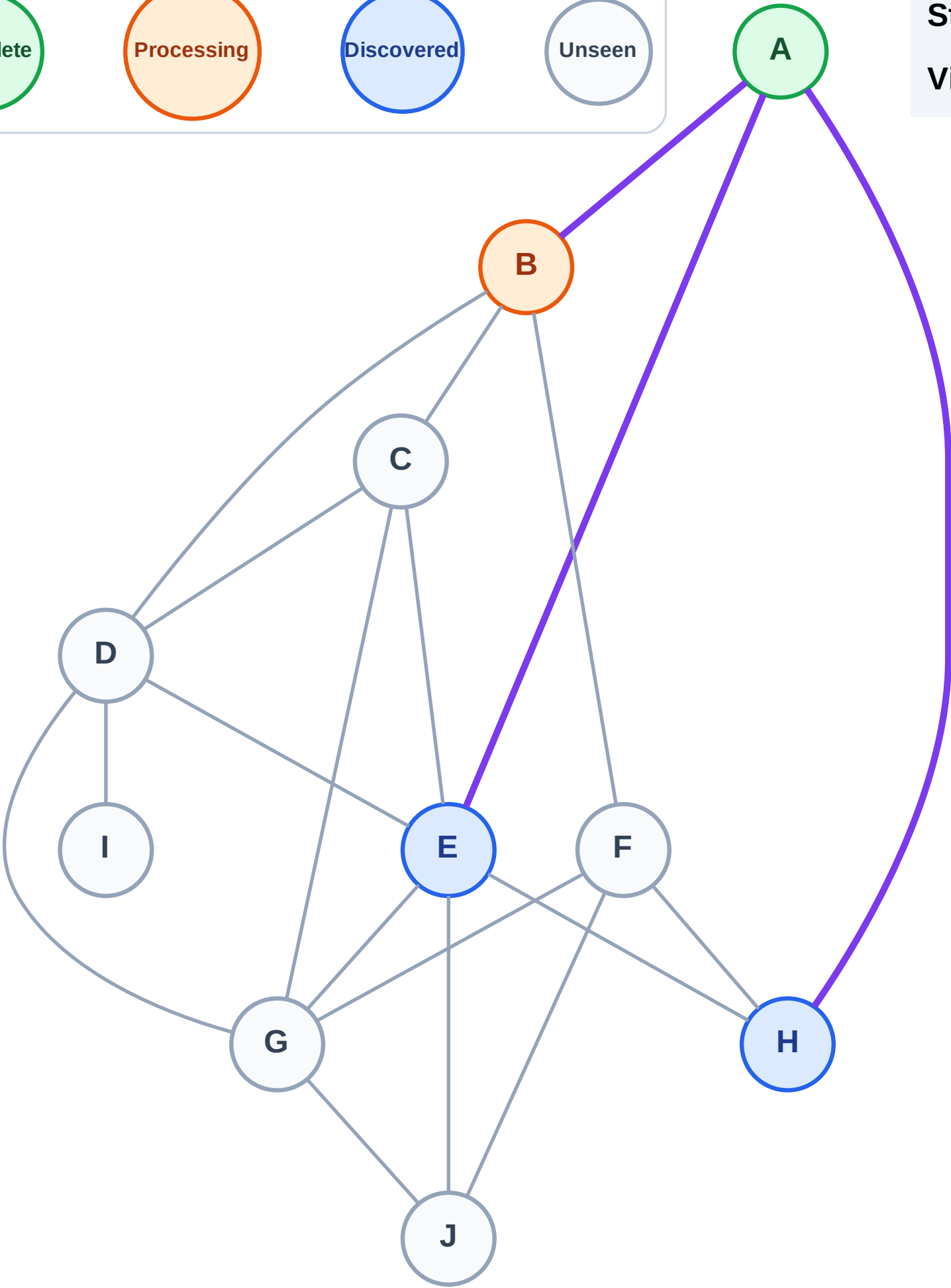
Processing

Discovered

Unseen

Stack (top at right): H | E

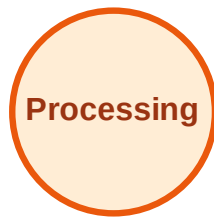
Visited: A



# DFS Traversal

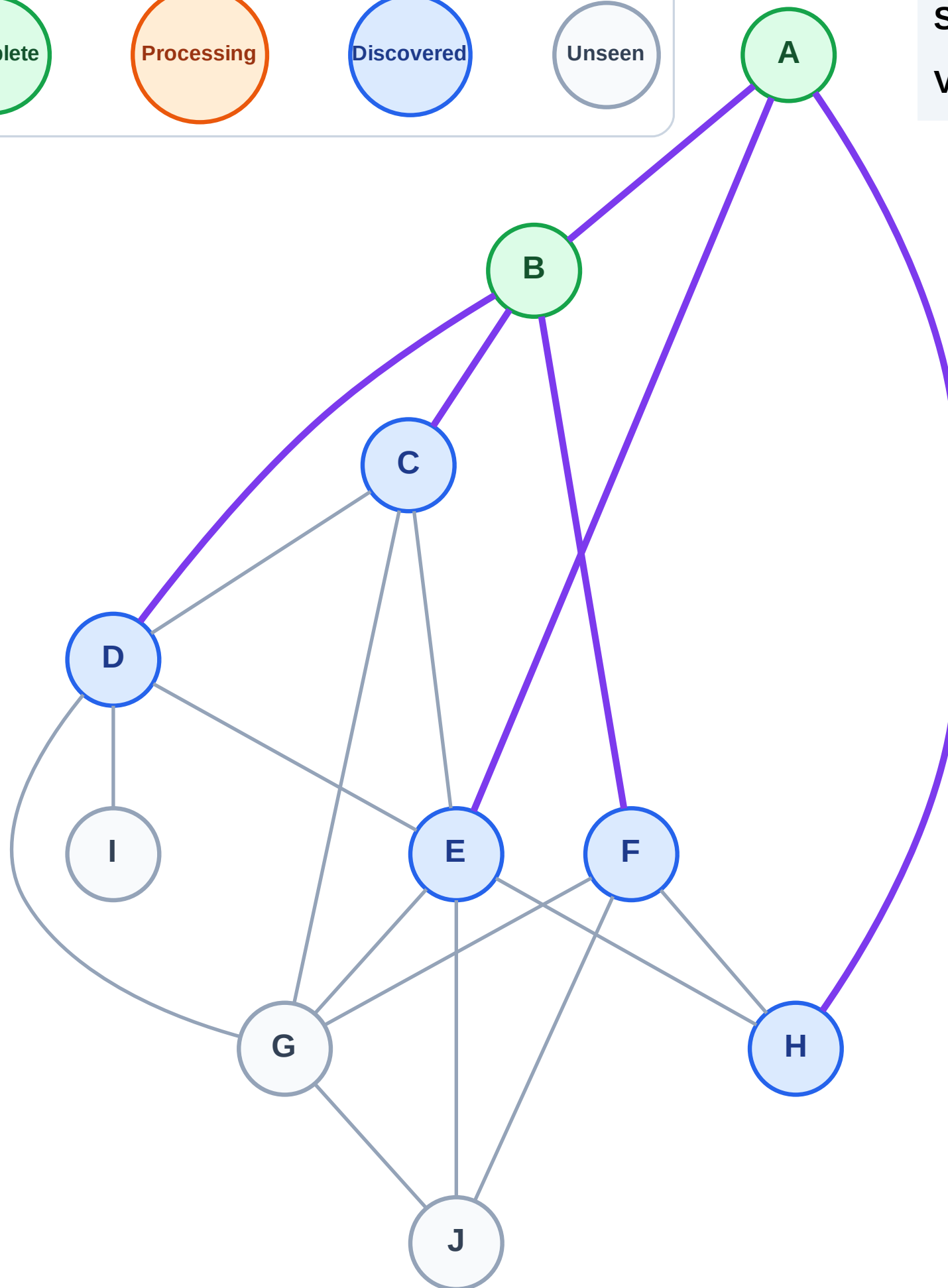
Step 5: Discover F, D, C from B

## Legend



Stack (top at right): H | E | F | D | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

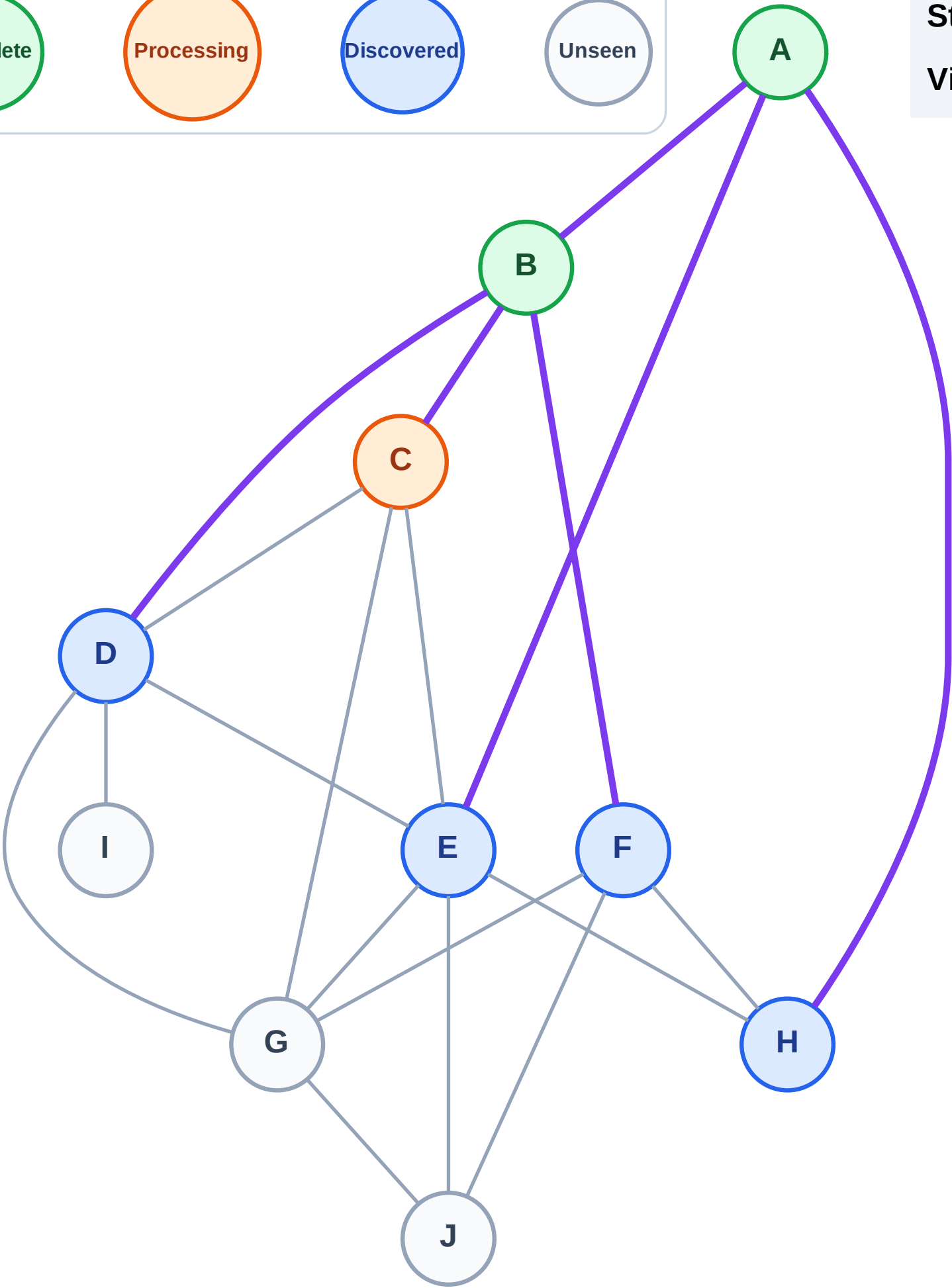
Processing

Discovered

Unseen

Stack (top at right): H | E | F | D

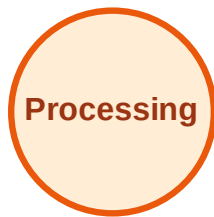
Visited: A, B



# DFS Traversal

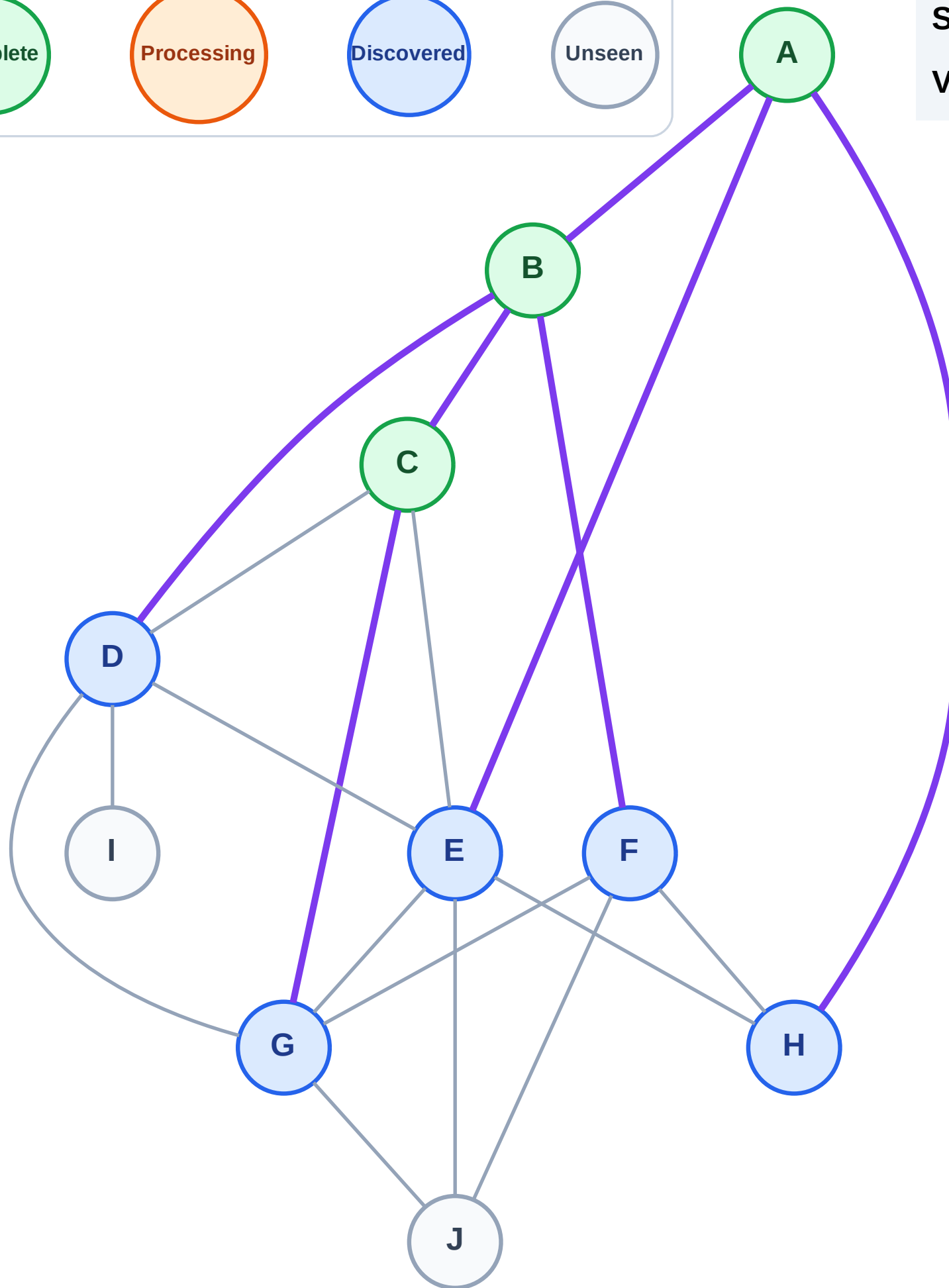
Step 7: Discover G from C

## Legend



Stack (top at right): H | E | F | D | G

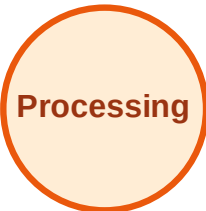
Visited: A, B, C



# DFS Traversal

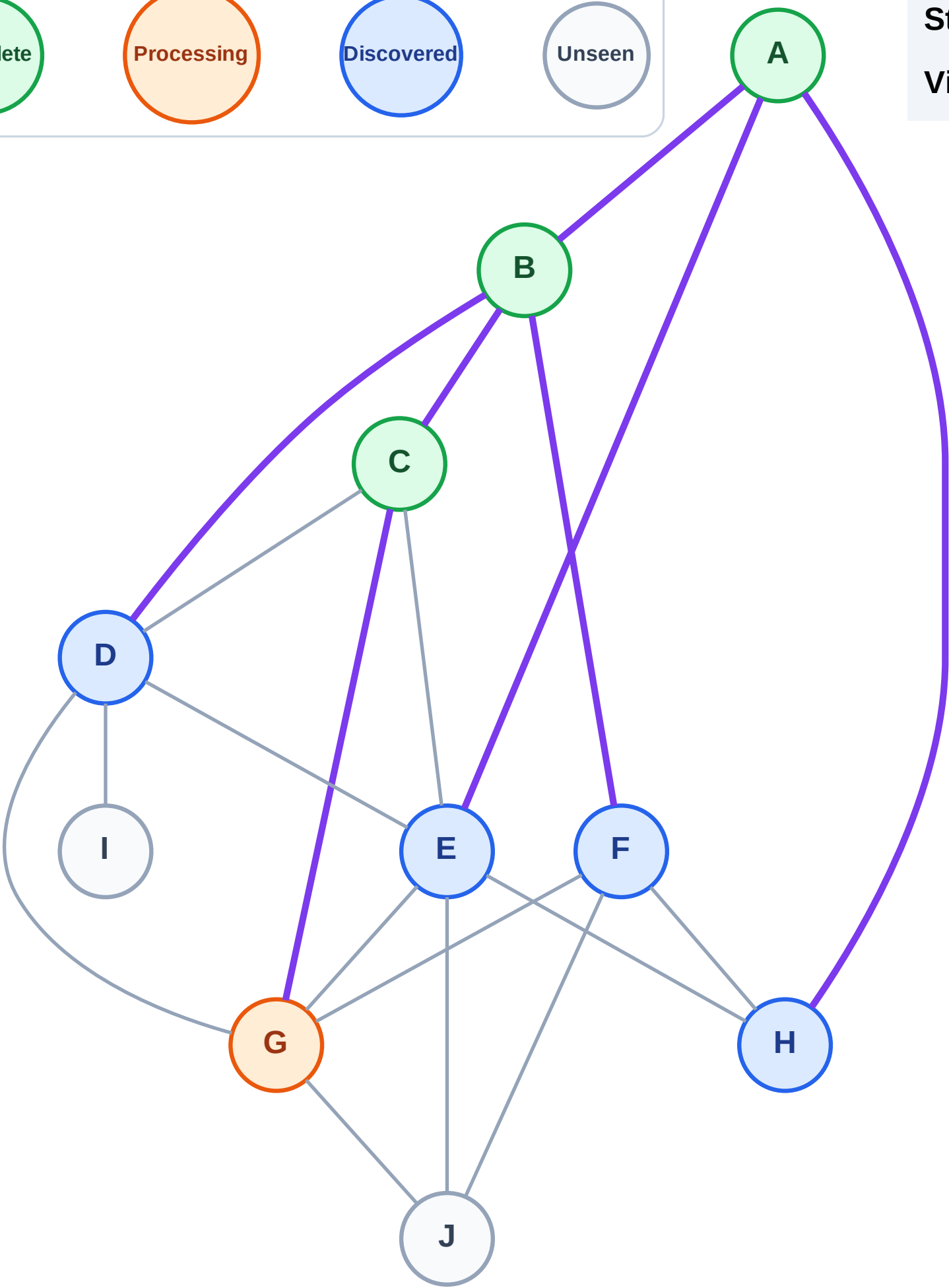
Step 8: Process node G

## Legend



Stack (top at right): H | E | F | D

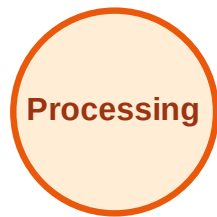
Visited: A, B, C



# DFS Traversal

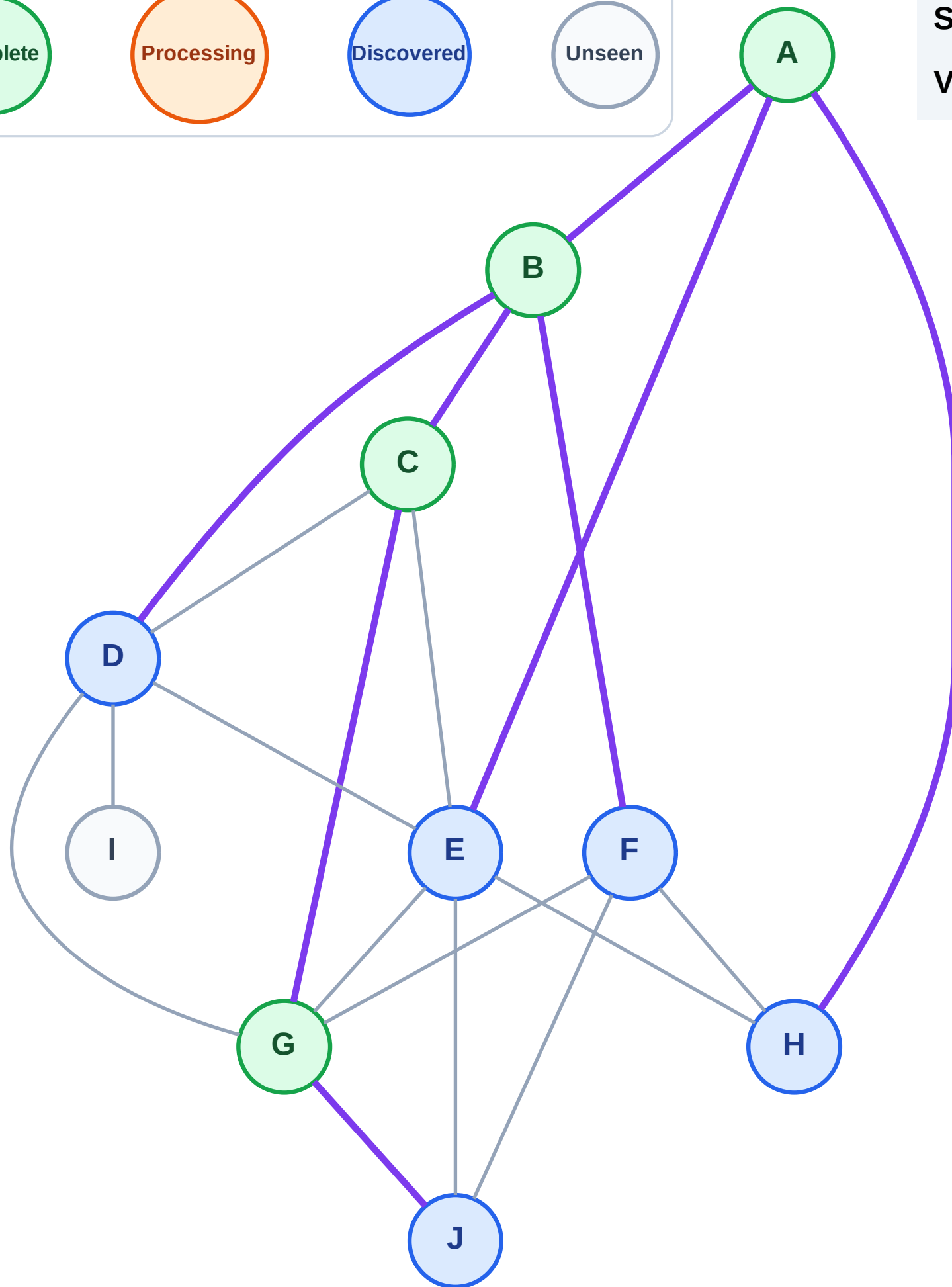
Step 9: Discover J from G

## Legend



Stack (top at right): H | E | F | D | J

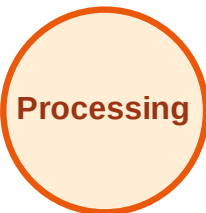
Visited: A, B, C, G



# DFS Traversal

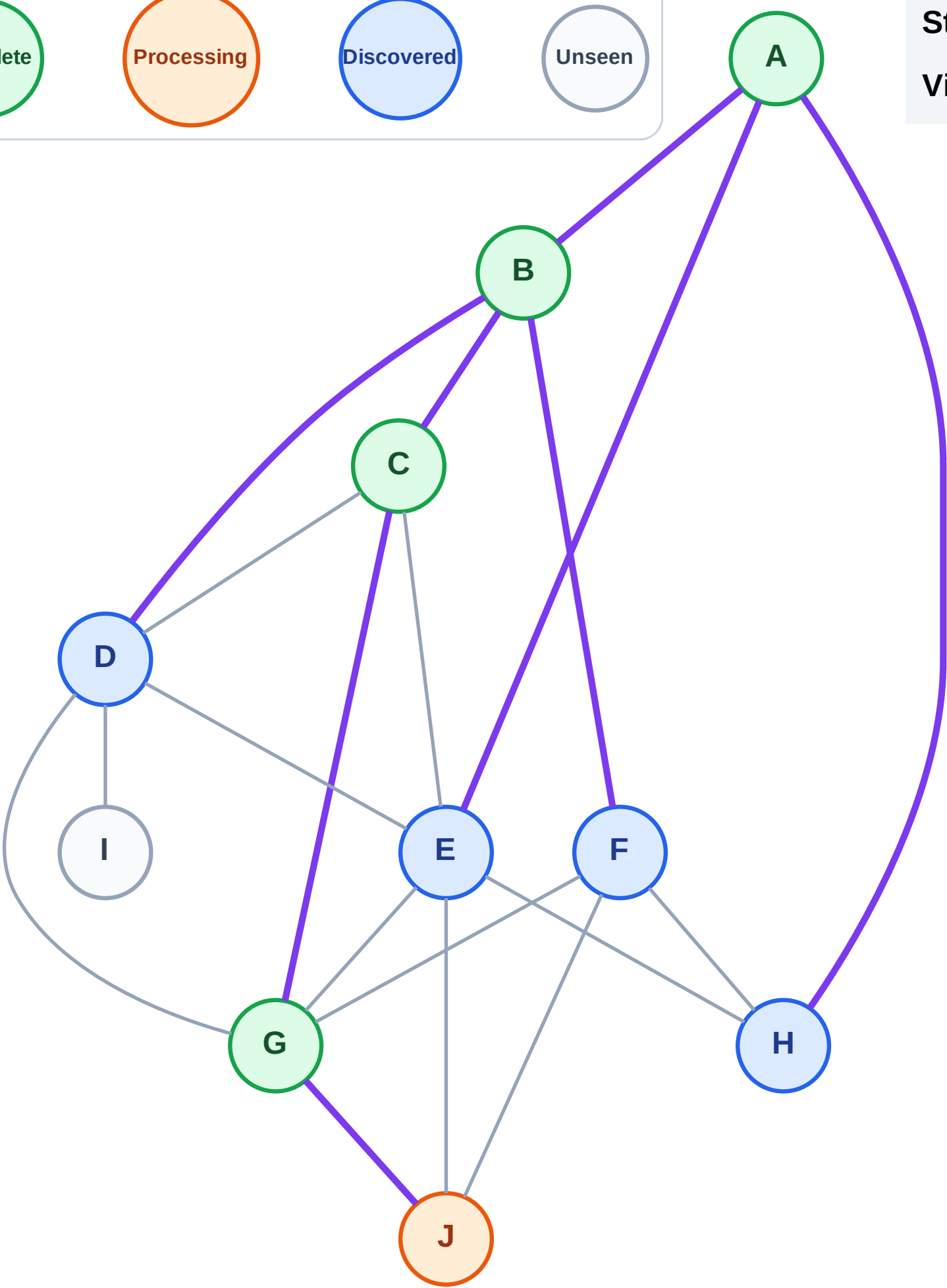
Step 10: Process node J

## Legend



Stack (top at right): H | E | F | D

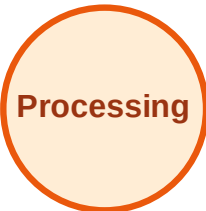
Visited: A, B, C, G



# DFS Traversal

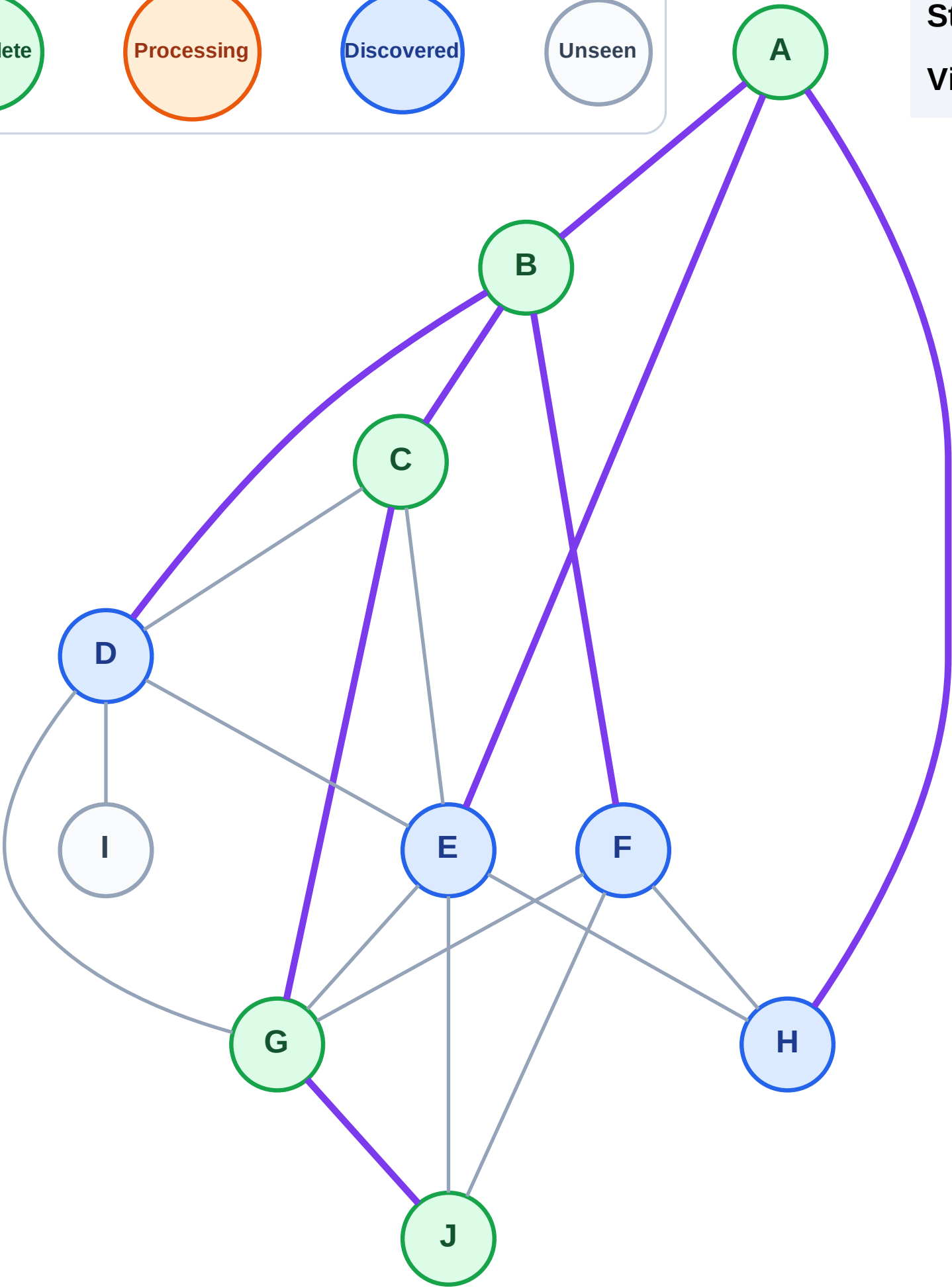
Step 11: No new neighbors from J

## Legend



Stack (top at right): H | E | F | D

Visited: A, B, C, G, J



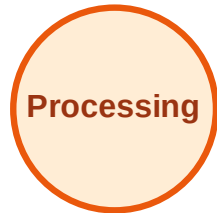
# DFS Traversal

Step 12: Process node D

## Legend



Complete



Processing



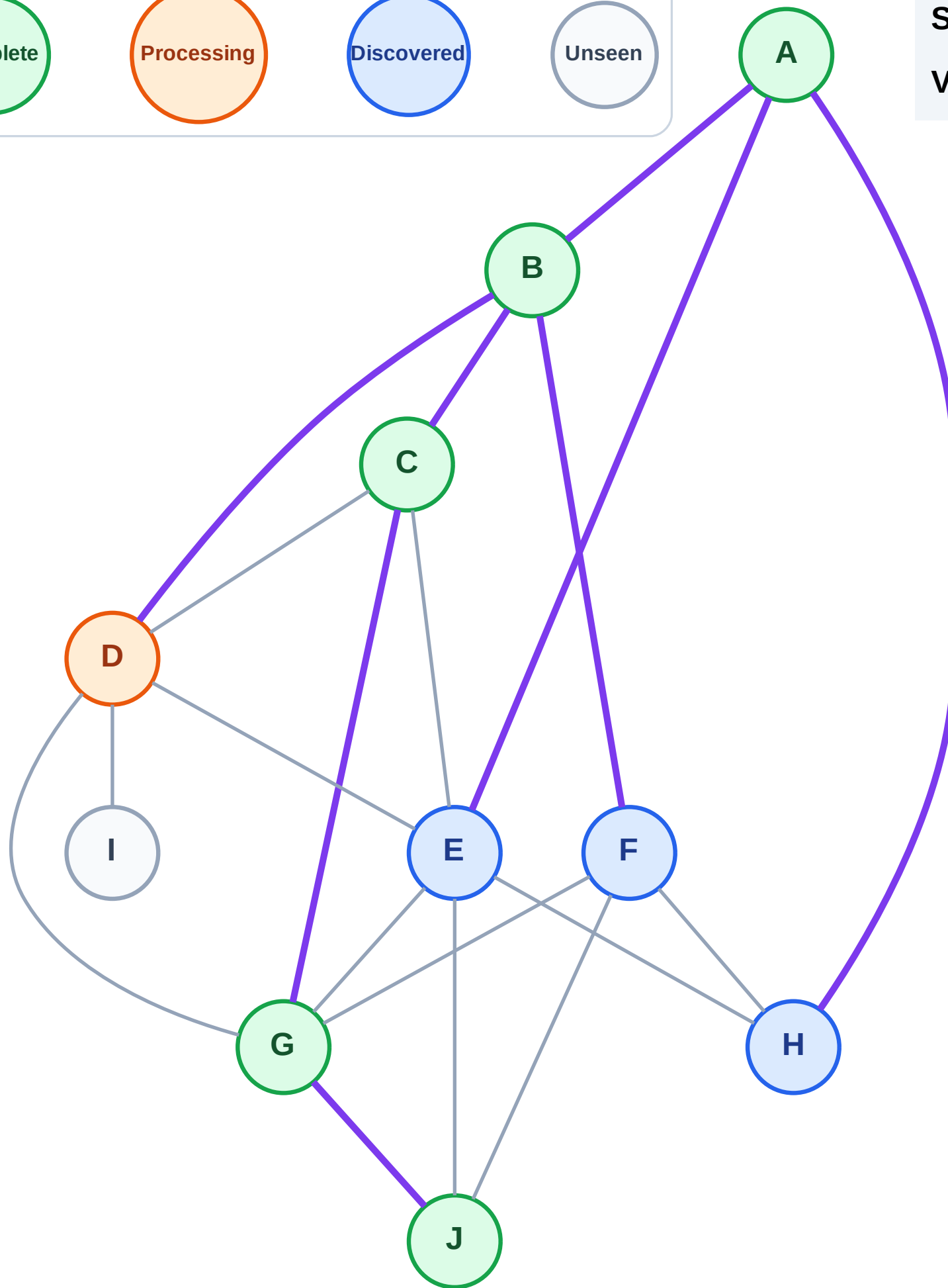
Discovered



Unseen

Stack (top at right): H | E | F

Visited: A, B, C, G, J



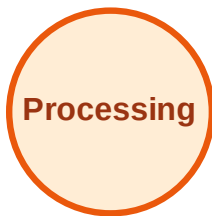
# DFS Traversal

Step 13: Discover I from D

## Legend



Complete



Processing



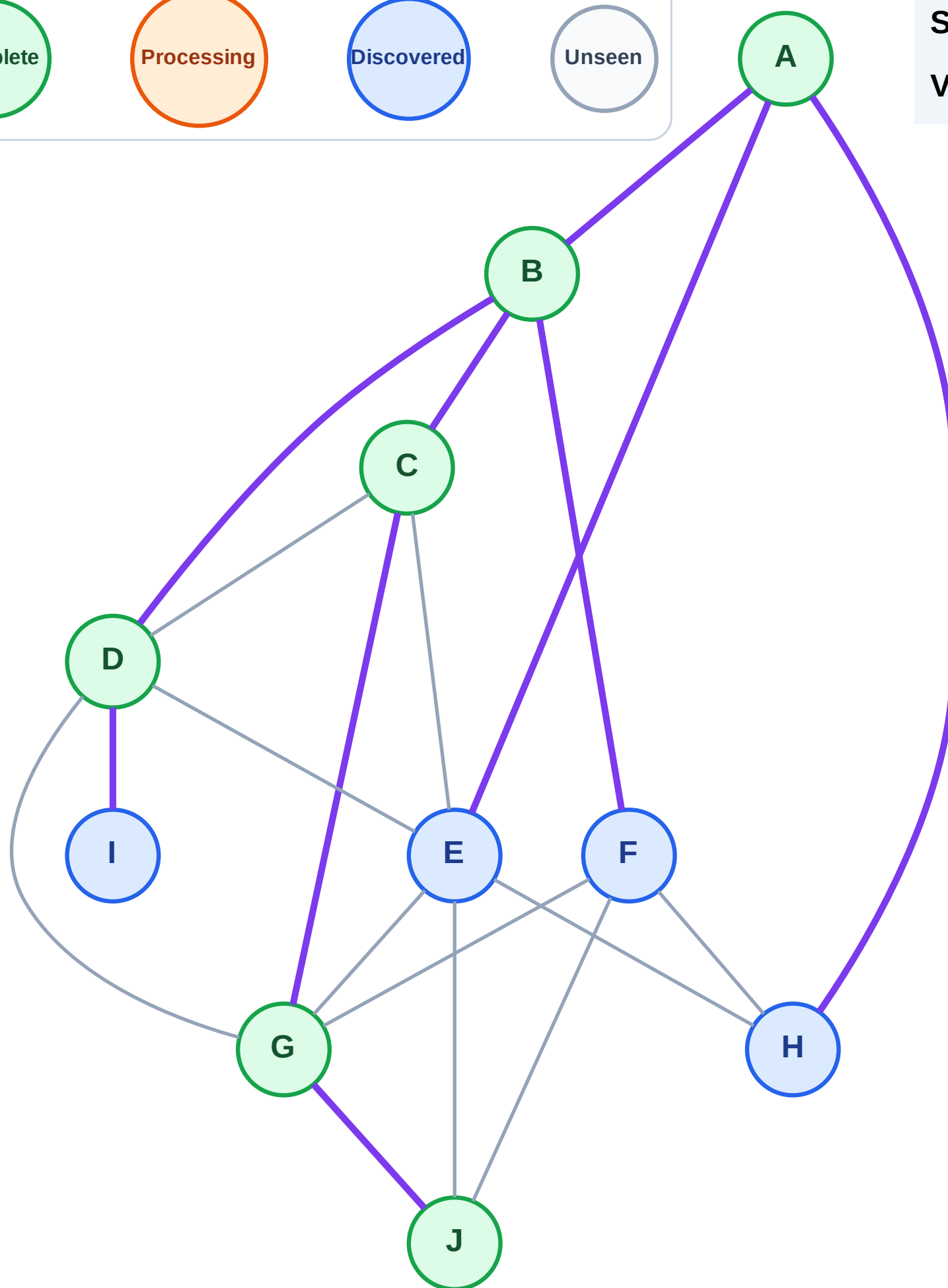
Discovered



Unseen

Stack (top at right): H | E | F | I

Visited: A, B, C, D, G, J



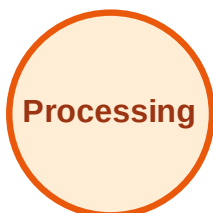
# DFS Traversal

Step 14: Process node I

## Legend



Complete



Processing



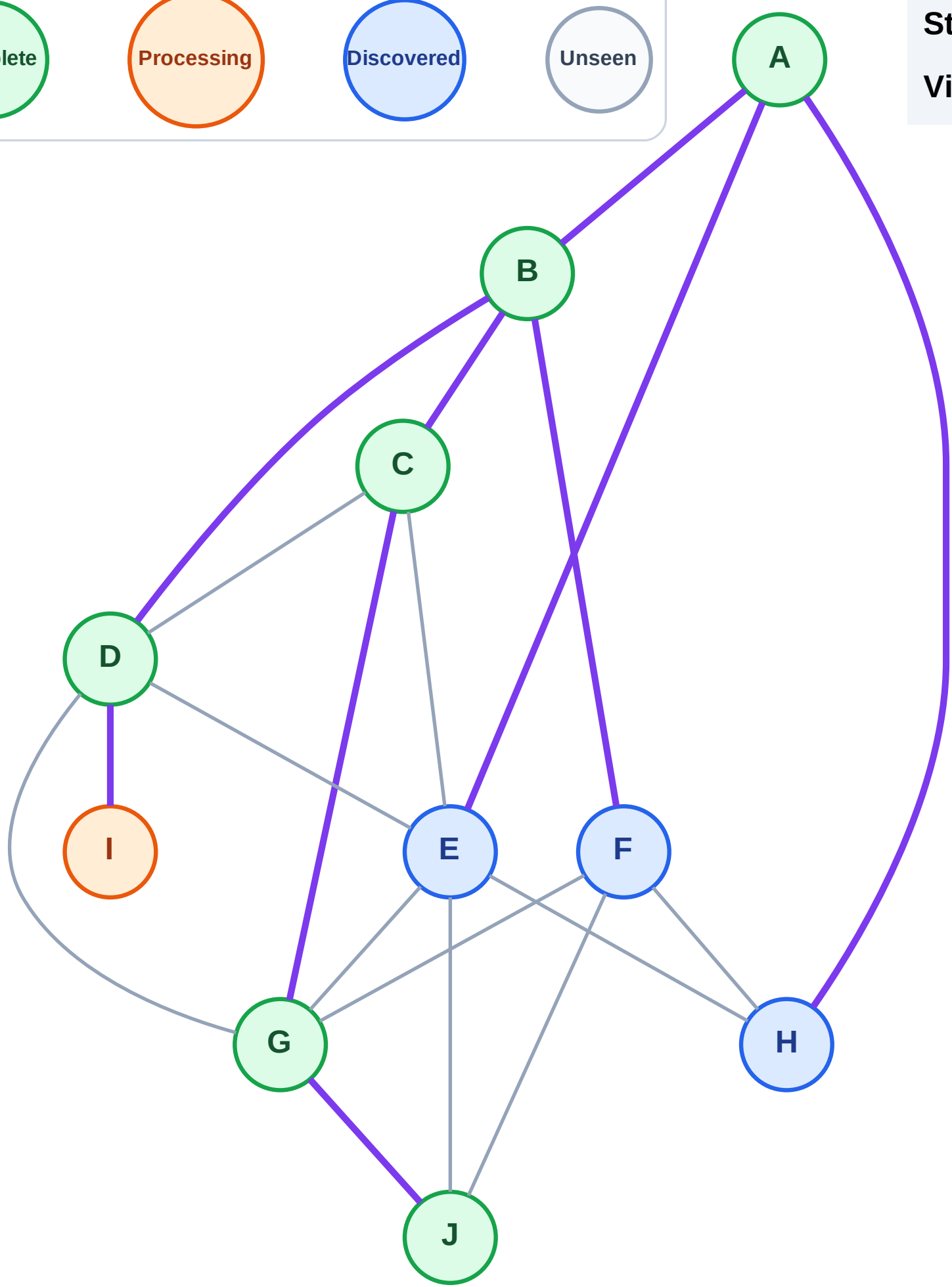
Discovered



Unseen

Stack (top at right): H | E | F

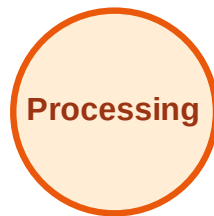
Visited: A, B, C, D, G, J



# DFS Traversal

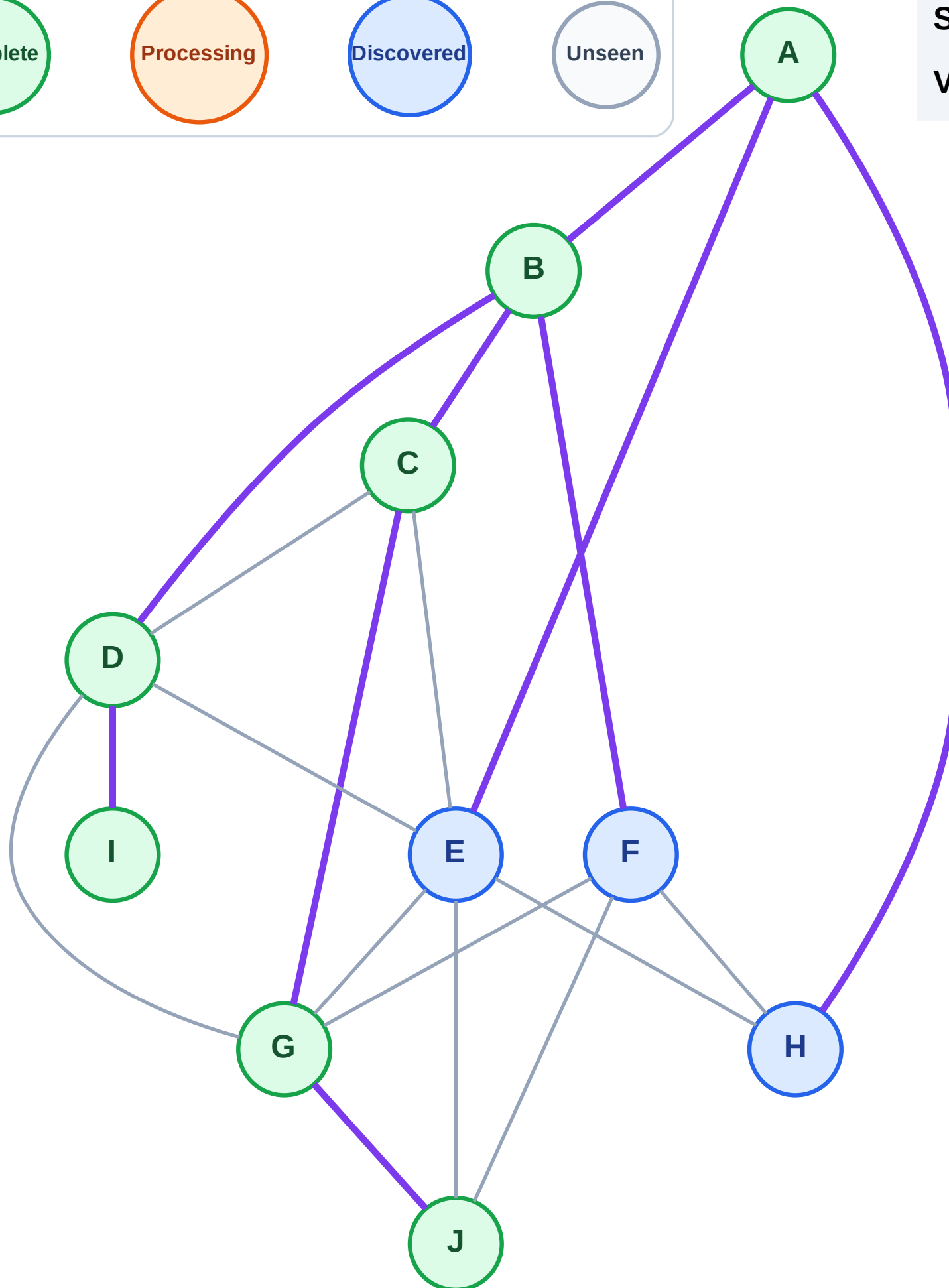
Step 15: No new neighbors from I

## Legend



Stack (top at right): H | E | F

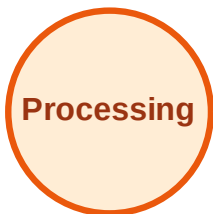
Visited: A, B, C, D, G, I, J



# DFS Traversal

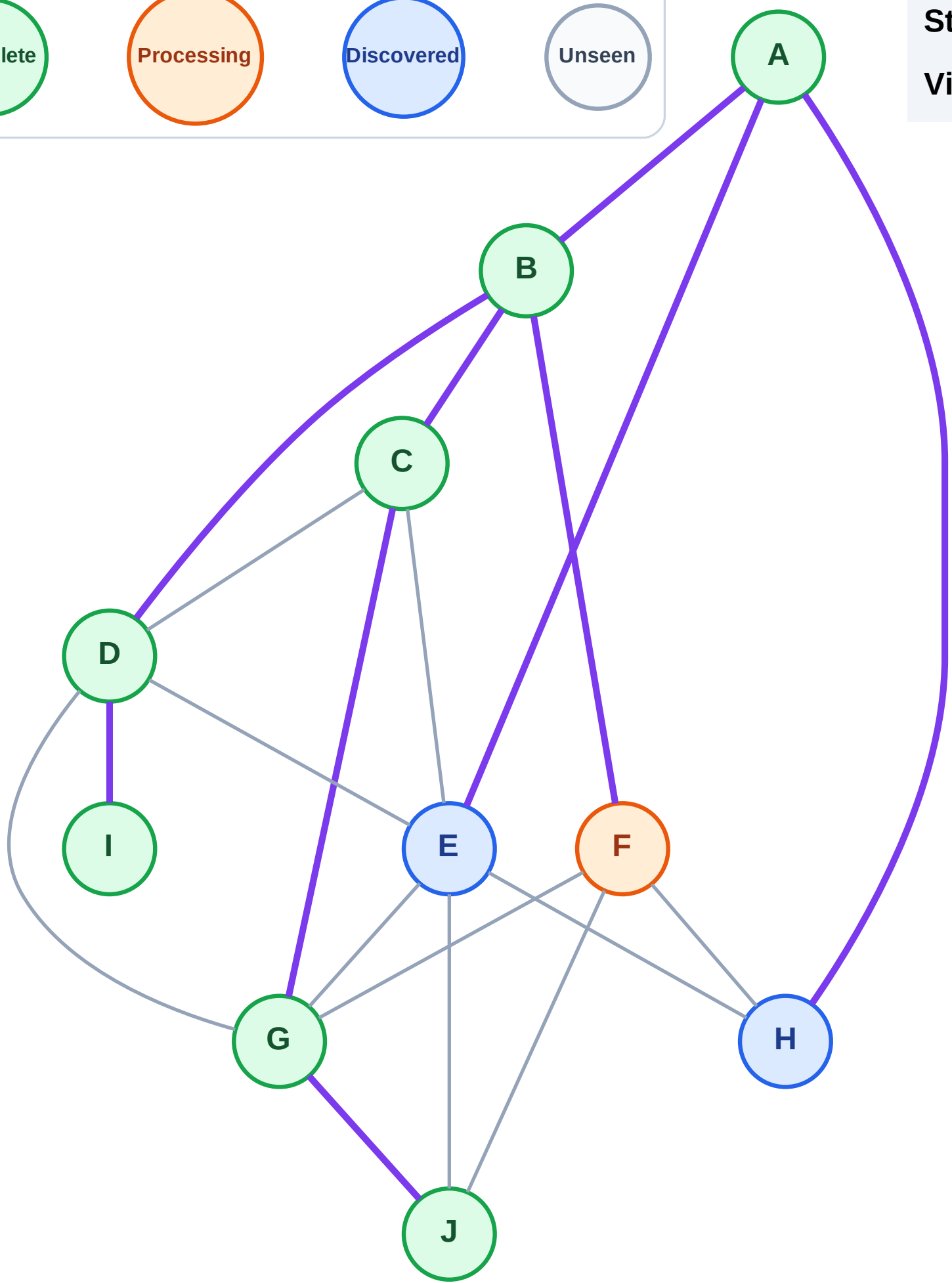
Step 16: Process node F

## Legend



Stack (top at right): H | E

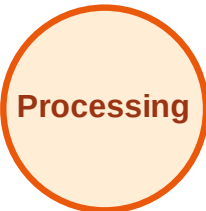
Visited: A, B, C, D, G, I, J



# DFS Traversal

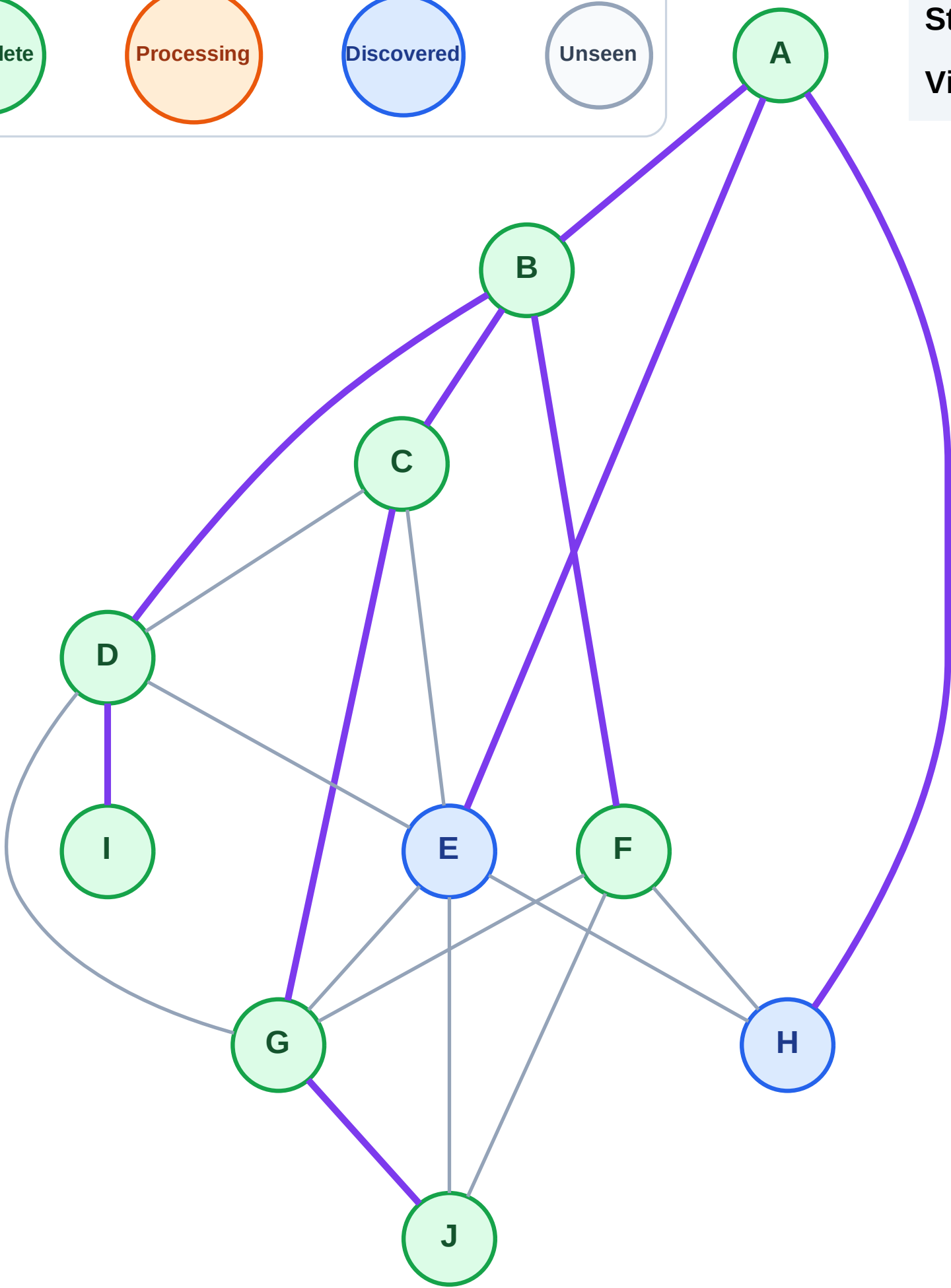
Step 17: No new neighbors from F

## Legend



Stack (top at right): H | E

Visited: A, B, C, D, F, G, I, J



# DFS Traversal

Step 18: Process node E

## Legend

Complete

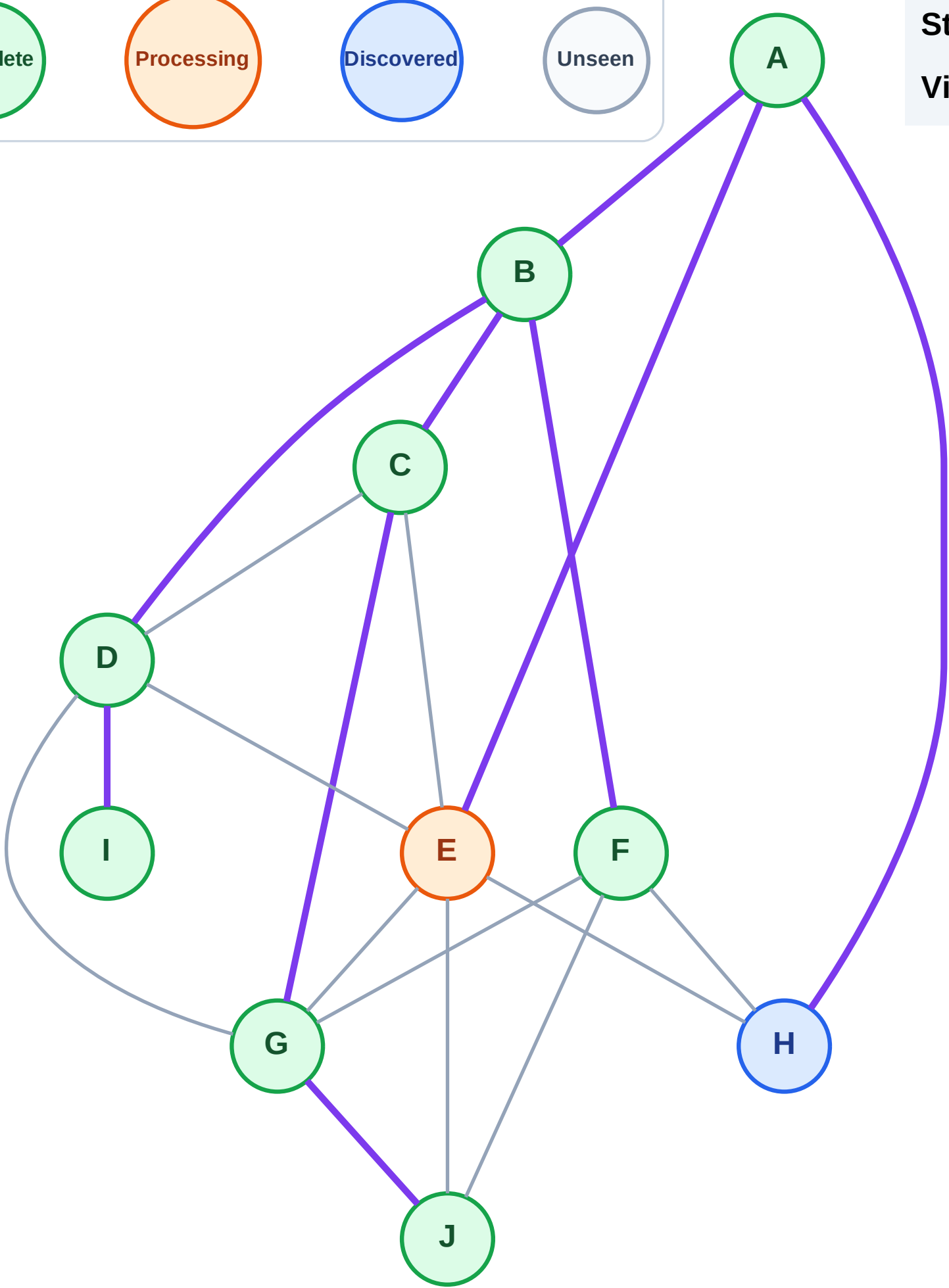
Processing

Discovered

Unseen

Stack (top at right): H

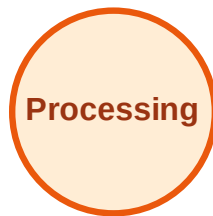
Visited: A, B, C, D, F, G, I, J



# DFS Traversal

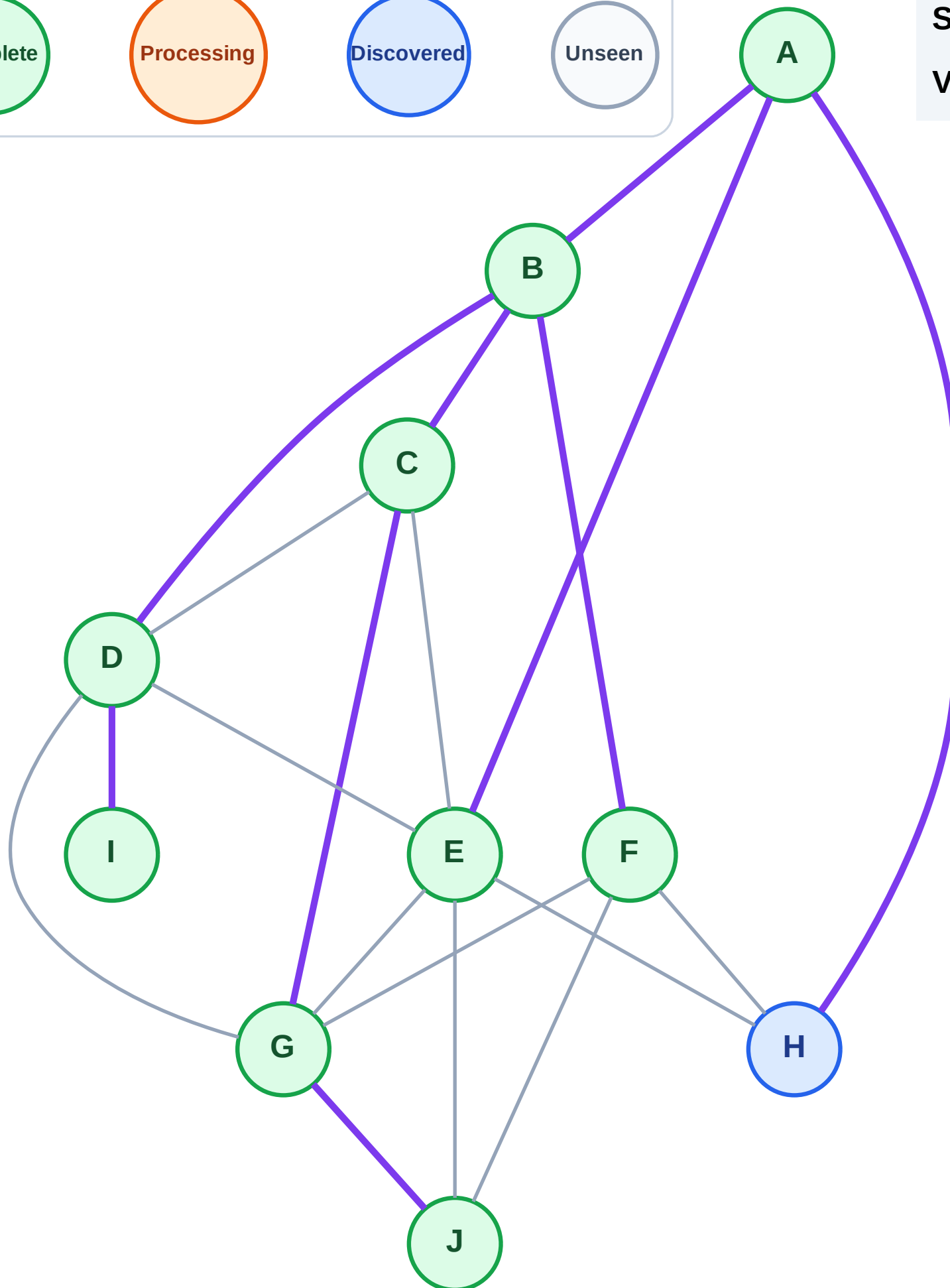
Step 19: No new neighbors from E

## Legend



Stack (top at right): H

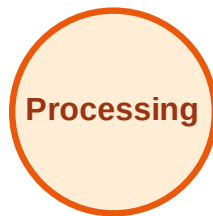
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

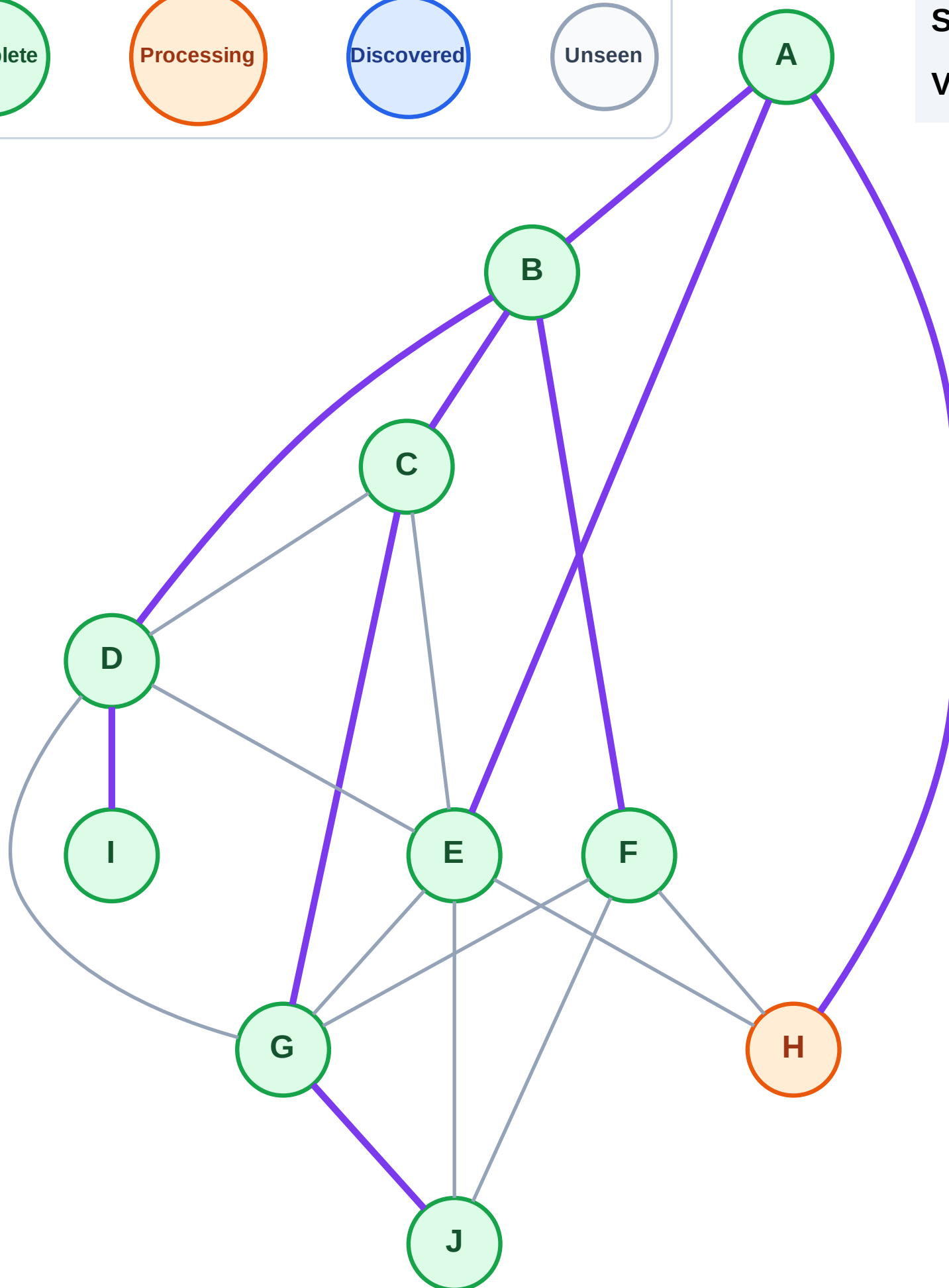
Step 20: Process node H

## Legend



Stack (top at right): (empty)

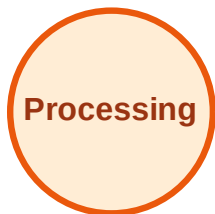
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

Step 21: No new neighbors from H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

